Causal Exaggeration: Unconfounded but Inflated Causal Estimates

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Abstract

The credibility revolution has made causal inference methods ubiquitous in economics. Yet there is widespread evidence of selection on significance and associated biases in the literature. I show that these two phenomena interact in ways that reduce the reliability of published estimates: while causal identification strategies alleviate bias from confounders, they reduce statistical power and can exacerbate another bias–exaggeration–when combined with selection on significance. I characterize this confounding-exaggeration trade-off theoretically and via realistic Monte Carlo simulations, and explore its empirical relevance in the literature. I then propose practical solutions, including a tool to identify the variation actually driving identification.

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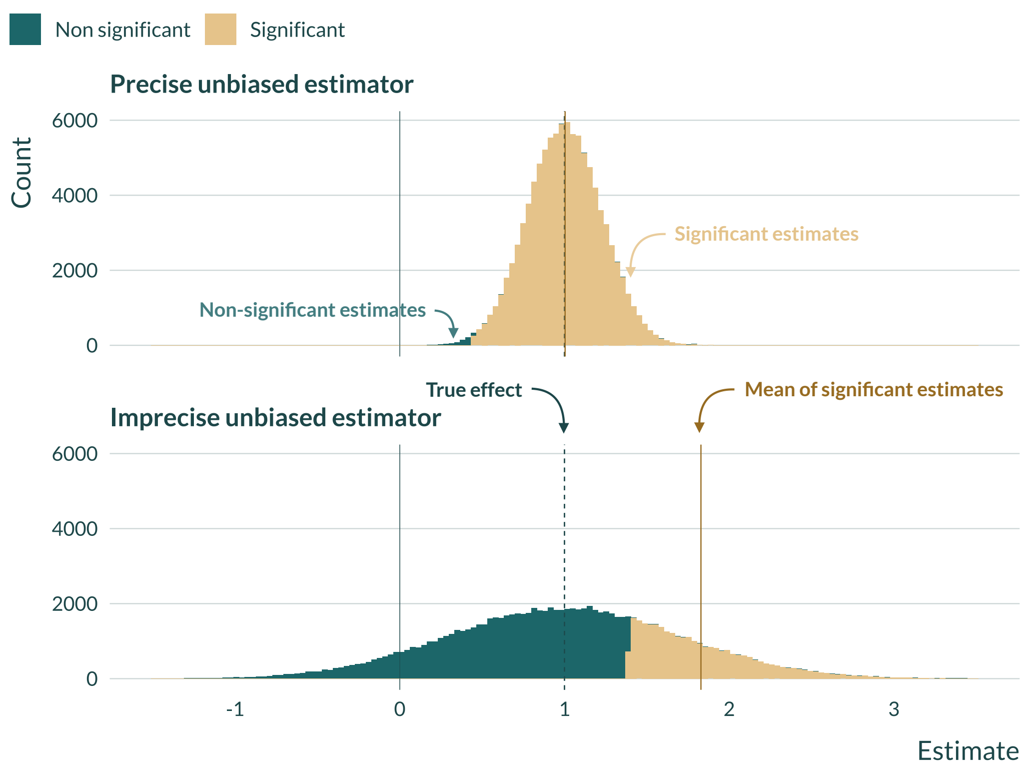
I Introduction

Identifying causal effects is one of the main objectives of empirical economics. Strategies such as Regression Discontinuity (RD), Instrumental Variables (IV), Difference-in-Differences (DiD), and event-study designs allow us to achieve this goal. To isolate causal effects, they only use a portion of the variation in the data; they focus on the exogenous part of treatment variation or restrict the sample to observations for which the random assignment assumption is credible. Limiting the variation used for identification can decrease precision and thus statistical power—the probability of rejecting the null hypothesis when it is false, or put simply, the probability of obtaining a statistically significant estimate. There is, therefore, a tension between reducing confounding and statistical power. Yet when statistical power is low, not only is the estimator imprecise but statistically significant estimates exaggerate the true effect size (Gelman and Tuerlinckx 2000, Ioannidis 2008, Gelman and Carlin 2014). I argue that for significant estimates, this creates a tension between reducing confounding and exaggerating the true effect size. While causal identification strategies alleviate confounding bias, they can introduce another form of bias—exaggeration—by limiting the variation used for identification.

A large literature has raised substantial concerns about biases resulting from confounding on the one hand and from publication bias and low power on the other (Ioannidis, Stanley and Doucouliagos 2017, Andrews and Kasy 2019, DellaVigna and Linos 2022, Young 2022, for instance). The present paper explores the interactions between approaches to address these two biases. It demonstrates the existence and practical importance of a confounding-exaggeration trade-off and studies its drivers. Building on prior literature reviews, I first examine exaggeration in the causal literature and discuss whether it can partly result from the loss in precision caused by the use of causal identification strategies. I then derive a formal proof of the confounding-exaggeration trade-off for prevailing identification strategies. I show that causal approaches can generate an exaggeration bias that exceeds the bias induced by confounders and describe the mechanisms through which it arises. Using calibrated Monte Carlo simulations, I further examine this causal exaggeration and its drivers in more complex concrete settings. Finally, I propose practical solutions to navigate the trade-off, including a tool to identify its main driver: the variation and observations that actually drive identification. While convincing identification strategies are essential to describe causal relationships, this paper emphasizes that improving identification can paradoxically pull estimates away from the true effect.

To clarify the trade-off and motivate its relevance, I next provide an overview of its component that has received relatively less attention in economics: exaggeration. Exaggeration of significant estimates arises in imprecise studies because such estimates constitute a selected subsample of all estimates (Gelman and Tuerlinckx 2000). Figure 1 illustrates this phenomenon. Estimates that are statistically significant at the 5% level are at least 1.96 standard errors away from zero; when standard errors are large, they lie in the tails of the distribution of all possible estimates and thus exaggerate the true effect. Power therefore matters even after obtaining a statistically significant estimate: when it is low, significant estimates from an unbiased estimator are necessarily far from the true effect, and the less precise the estimator, the larger the exaggeration. An estimator $\hat{\beta}$ of the true effect β might be unbiased in the traditional sense of $\mathbb{E}[\hat{\beta}] = \beta$ but conditionally biased in the sense that $\mathbb{E}[\hat{\beta}|\text{ Significant}] \neq \beta$.

Figure 1 – Distribution of Two Unbiased Estimators with Different Variances and Significance of Resulting Estimates



Notes: 100,000 draws from two normal distributions $\mathcal{N}(1, 0.05)$ and $\mathcal{N}(1, 0.5)$. Significance level: 5%

Exaggeration of published estimates occurs when two conditions arise simultaneously: 1) publication bias favors statistically significant results, and 2) statistical power is low. Both features are observed in the economics literature. First, a large body of literature documents selection on statistical significance in economics (Rosenthal 1979, Brodeur et al. 2016, Andrews and Kasy 2019, Vivalt 2019, Abadie 2020, Brodeur, Cook and Heyes 2020, for instance). Second, additional studies have highlighted a frequent and substantial lack of statistical power and resulting exaggeration (Ioannidis, Stanley and Doucouliagos 2017, Ferraro and Shukla 2020, Young 2022, Bagilet 2023). Even in experimental economics, with a high level of control and an arguable absence of confounders, laboratory experiments published in top economics journals failed to replicate, with original estimates inflated by a factor of at least 1.5 on average (Camerer et al. 2016). In the field, DellaVigna and Linos (2022) finds that results from nudge experiments published in academic journals are about 6 times larger than those from large-scale experiments and explains this discrepancy by a lack of power, selection on significance, and characteristics of the experiments. In the non-experimental economics literature, where statistical power is rarely a central consideration once a significant estimate has been obtained, several meta-analyses point to consequential exaggeration. Ioannidis, Stanley and Doucouliagos (2017) estimates the median statistical power across a wide range of areas in economics at only 18%. Despite typically large sample sizes in economics, the paper documents substantial exaggeration in the literature: nearly 80% of estimates are likely exaggerated by a factor of two. In addition, an expert survey in political science–a field with comparable power issues–suggests that applied researchers largely underestimate the prevalence of these issues (Arel-Bundock et al. 2024). This body of evidence establishes that exaggeration is both widespread and substantial in economics. Taking exaggeration into account and understanding its drivers is therefore critical, particularly when published estimates inform policy decisions. Yet while exaggeration itself is well documented, its drivers remain understudied.

The main insight of the present paper is that the use of causal identification strategies can cause exaggeration by limiting the variation used for identification. While each strategy does so through a distinctive channel, they all discard variation. RD designs only consider observations within the bandwidth, decreasing the effective sample size and thus precision. An IV strategy only uses the subset of the variation in the treatment that is explained by the instrument. Designs leveraging exogenous shocks may identify effects from only a small set of treated observations. Matching prunes units that cannot be matched and thus reduces the effective sample size. As illustrated below, adding controls or fixed effects can increase the variance of the estimator if they absorb more of the variation in the treatment than in the outcome variable. Since causal identification strategies can be interpreted as ways of controlling for confounders, [1] this last point ties all the strategy-specific mechanisms together and leads to a central contribution of the paper: showing that when identification strategies absorb more of the variation in the treatment than in the outcome, they increase the variance of the resulting estimator and can cause exaggeration. Considering a simple linear homoskedastic model gives the intuition. Let $y_i = \alpha + \beta x_i + \delta w_i + u_i$, $\forall i \in \{1, .., n\}$, with x the variable of interest, w a potentially unobserved variable correlated with x and u, an error term. Under usual assumptions and using the Frisch-Waugh-Lovell theorem, we

1. Fixed Effects (FEs) based identification strategies such as DiD control for the invariant, unobserved, and arguably endogenous part of the variation in the outcome. An IV approach essentially partials out the variation in x unexplained by the instruments. Fuzzy-RD and propensity score matching can be thought of as control function approaches, of the forcing variable and propensity score respectively. In addition, excluding observations that are outside the bandwidth or unmatched is equivalent to controlling for observation-level fixed effects for these units.

get that $\mathbb{V}_{\hat{\beta}_{\text{OMIT}}}$ and $\mathbb{V}_{\hat{\beta}_{\text{CTRL}}}$, the variance of the estimators for β when omitting w and controlling for it are respectively:

$$\mathbb{V}_{\hat{\beta}_{\text{OMIT}}} = \frac{\sigma^2_{u_{\text{OMIT}}}}{n\ \sigma^2_x} = \frac{\sigma^2_{y^{\perp x}}}{n\ \sigma^2_x} \qquad \text{and} \qquad \mathbb{V}_{\hat{\beta}_{\text{CTRL}}} = \frac{\sigma^2_{u_{\text{CTRL}}}}{n\ \sigma^2_{x^{\perp w}}} = \frac{\sigma^2_{y^{\perp x,w}}}{n\ \sigma^2_{x^{\perp w}}}$$

where $\sigma^2_{u_{\text{OMIT}}}$ and $\sigma^2_{u_{\text{CTRL}}}$ are the variances of the residuals in the regression of y on x and of y on x and w respectively or equivalently the variances of the parts of y that are orthogonal to x and to x and w respectively ($\sigma^2_{y^{\perp x}}$ and $\sigma^2_{y^{\perp x,w}}$), σ^2_x is the variance of x and $\sigma^2_{x^{\perp w}}$ is the variance of the part of x orthogonal to w. Thus,

$$\mathbb{V}_{\hat{\beta}_{\text{OMIT}}} < \mathbb{V}_{\hat{\beta}_{\text{CTRL}}} \quad \Leftrightarrow \quad \frac{\sigma^2_{y^{\perp x}}}{n\ \sigma^2_x} < \frac{\sigma^2_{y^{\perp x,w}}}{n\ \sigma^2_{x^{\perp w}}} \quad \Leftrightarrow \quad \frac{\sigma^2_{x^{\perp w}}}{\sigma^2_x} < \frac{\sigma^2_{y^{\perp x,w}}}{\sigma^2_{y^{\perp x}}}$$

Controlling for w increases the variance of the estimator if the fraction of the variance unexplained by w is greater for $y^{\perp x}$ than for x. Put differently, if controlling absorbs more of the variation in x than in the residual part of y ($y^{\perp x}$), the variance of the estimator increases. Since exaggeration increases with estimator's variance, controlling for a confounder can increase exaggeration. Controls, including fixed effects, limit the identifying variation and, as a result, observations whose treatment status is largely explained by these controls contribute little, if at all, to identification. Identification may thus effectively rely on a small sample. Under some circumstances discussed in Sections III and IV, controlling can even produce an exaggeration bias larger than the Omitted Variable Bias (OVB) that would result from an absence of controls.

In the remainder of the paper, I first document the magnitude of the trade-off. To do so, I build on existing literature reviews (Brodeur, Cook and Heyes 2020, Young 2022, Lal et al. 2024) to discuss evidence of exaggeration bias in a large set of causal studies mostly published in top journals. These analyses reveal heterogeneity across studies: while exaggeration might be limited in some analyses, it is likely substantial in many others. For instance, in the set of papers reviewed in Young (2022), I show that half of the IV designs would exaggerate a true effect size of the magnitude of the "naive" OLS estimate by a factor larger than 2.4—and by a factor larger than 2 for headline results. I then directly compare confounding and exaggeration biases for an example study. I show that in this instance the bias of the IV could be on par or even larger than that of the OLS and that exaggeration may explain this difference.

Next, I derive a formal proof of the existence of the trade-off for prevailing causal identification strategies. Specifically, I show that the bias caused by exaggeration can be larger than the one caused by confounders. I also analyze the drivers of exaggeration and demonstrate that it increases as the strength of the instruments decreases, the number of exogenous shocks decreases or when controlling for a confounder absorbs more of the variation in the treatment than in the outcome.

Then, I further investigate exaggeration and its determinants in concrete settings where no closed-form expressions are available. Because exaggeration is defined relative to the true effect, which is never observed in practice, I rely on simulations to quantify it. Monte Carlo simulations also allow me to vary parameters of interest *ceteris paribus*. To make these simulations concrete, I calibrate them to emulate existing studies from education, environmental, health, labor, and political economics. When the variation remaining for identification is limited, exaggeration can be substantial: in these calibrated simulations, it ranges from 1.3 to 4 for realistic parameter choices.

Finally, I discuss concrete avenues to address causal exaggeration when carrying out a non-experimental study.[2] A series of tools can be used to evaluate the potential magnitude of confounding and exaggeration issues separately. Sensitivity analyses help with the former while power calculations help with the latter. Considering the attention given to bias avoidance in the economics literature, making power central to non-experimental analyses, even after a significant estimate has been obtained, would limit bias caused by exaggeration. Prospective power simulations help identify the design parameters affecting power and exaggeration by approximating the data generating process (Gelman 2020, Black et al. 2022). Retrospective power calculations allow evaluating whether a study would have had enough power to confidently estimate a range of smaller but credible effect sizes (Gelman and Carlin 2014, Stommes, Aronow and Sävje 2023). Focusing more specifically on the drivers of the trade-off, I introduce an `R` package, `ididvar`, that facilitates the identification of the variation underlying causal estimates. The package provides tools to compute each observation's contribution to the estimated effect and to characterize the effective sample used for identification.

This paper contributes to four strands of the applied economics literature. First, the idea that causal identification estimators, while unbiased, may be imprecise is not new; it is part of the well-known bias–variance trade-off (Imbens and Kalyanaraman 2012, Deaton and Cartwright 2018, Hernán and Robins 2020, Ravallion 2020). I approach this literature from a different angle; through the prism of statistical power and publication bias. Limited precision resulting from the use of causal identification strategies not only makes it difficult to draw clear conclusions about the exact magnitude of effects, but, I argue, can also inherently lead to inflated published estimates, creating another form of bias. In this sense, the bias–variance trade-off is in fact a bias–bias trade-off.

Second, this paper contributes to the literature on statistical power and exaggeration. As discussed above, a growing literature reveals that these issues are widespread and consequential in

2. In experimental studies, there are essentially no confoundings. A solution to increase power and reduce exaggeration is generally to increase sample size, reduce noise by improving measurement or improving balance or to focus on larger potential effects.

economics. However, studies in this literature rarely investigate the determinants or consider specific causal identification strategies separately (Ioannidis, Stanley and Doucouliagos 2017, Schell, Griffin and Morral 2018, Ferraro and Shukla 2020, Black et al. 2022, Stommes, Aronow and Sävje 2023, Arel-Bundock et al. 2024). To my knowledge, only a limited number of studies explore broader determinants: Vu (2025) analyzes the impact of standard error corrections and in a compagnon paper, I study concrete design parameters that can generate exaggeration across a wide range of settings (Bagilet 2023). In the present paper, I link the exaggeration literature to core causal inference principles and propose an overarching mechanism inherent to causal identification strategies as a whole: although each strategy does so through different means, they all discard part of the variation, thereby increasing the risk of exaggeration.

Third, this paper makes a methodological and practical contribution by facilitating the computation of concrete measures of each observation's contribution to identification and, by extension, of the effective sample underlying the estimated effect. It extends and complements existing approaches (Angrist and Pischke 2009, Aronow and Samii 2016) and facilitates their implementation and interpretability in applied economics research. The accompanying `R` package enables users to compute, visualize, and analyze these measures in a straightforward and consistent way. It allows applied researchers to diagnose potential exaggeration risks *ex-post*, without assumptions about true effect sizes. The package also provides a framework for identifying and communicating the set of observations contributing to identification and how this set changes across specifications, levels of fixed effects and sets of control variables.

Finally, this study contributes to the literature on replicability in economics (Camerer et al. 2016, Ioannidis, Stanley and Doucouliagos 2017, Christensen and Miguel 2018, Kasy 2021). The trade-off presented in the present paper suggests that the widespread reliance on convincing causal identification strategies in economics does not necessarily shield the field from replication threats. The tools discussed and introduced can be readily used in replication analyses or requested by referees to transparently document statistical power and the sources of identifying variation.

In the following section, I document evidence of causal exaggeration in the causal economics literature. In Section III, I study the drivers of exaggeration and formally show in a simple setting that the use of causal identification strategies can exacerbate it. In Section IV, I implement realistic Monte-Carlo simulations to illustrate the existence of the confounding-exaggeration trade-off. I discuss potential solutions to navigate this trade-off and introduce the `ididvar` package in Section V and conclude in Section VI.

II Causal Exaggeration in the Literature

A notable pattern in the economics literature is that causal methods routinely yield larger estimates than association-based approaches. The data in Young (2022) reveals that among 30 reproducible papers published in *American Economic Association* journals,[3] the median ratio of headline IV to corresponding OLS estimates is 2.3, with 25% of estimates exceeding a ratio of 5.4. A comparable phenomenon appears in top political science journals (Lal et al. 2024). In the literature on acute health effects of air pollution, which I explore in a companion paper, causal estimates are substantially larger than what would have been predicted by the standard epidemiology literature, with estimates regularly 10 times larger or more (Bagilet 2023). Even within individual studies, the median ratio of 2SLS to corresponding "naive" OLS estimates is 3.8.

What explains these substantially larger effect sizes from causal methods compared to non-causal approaches? While they might remove omitted variable bias, reduce attenuation bias from classical measurement error, or target different causal estimands, exaggeration and imprecision likely explain part of this difference. In the present section, I explore this possibility.

II. 1 Quantifying exaggeration

As discussed in the introduction, lack of power is widespread and exaggeration substantial in economics; Ioannidis, Stanley and Doucouliagos (2017) estimates that nearly 80% of results published in a wide array of empirical economic literatures are likely exaggerated, typically by a factor of two and one-third by a factor of four or more. Other studies corroborate this finding. Using a more conservative approach, Ferraro and Shukla (2020) estimates that 56% of results in environmental economics are exaggerated by a factor of two or more. Arel-Bundock et al. (2024) finds that estimates in the political science are about 3 times too large. DellaVigna and Linos (2022) shows that, due to exaggeration, academic papers on nudge experiments find results that are typically 6 times larger than in large-scale experiments. To complement this evidence, I assess whether comparable patterns of exaggeration could be present in causal studies specifically.

I begin by examining a broad literature that in return requires strong assumptions about true effect sizes. I leverage data from Brodeur, Cook and Heyes (2020), which reviews the universe of hypothesis tests reported in papers published in the top-25 economics journals in 2015 and 2018 using RCT, DID, RDD or IV. The paper shows that IV, and to a lesser extent, DiD are particularly subject to publication bias, one of the two ingredients of exaggeration. Since the exaggeration ratio is the expected value of the absolute value of significant estimates over the

3. The exact selection process is described in Section 3 of Young (2022). The sample consists of papers using the keyword "instrument", published through July 2016, that are reproducible, rely on open-access data and Stata code, and use linear methods with standard covariance estimates.

true effect, it can only be calculated by hypothesizing true effect sizes. Given the breadth and heterogeneity of papers considered, hypothetical effect sizes cannot be derived from theory or meta-analyses. To circumvent this limitation, I evaluate the proportion of studies in Brodeur, Cook and Heyes (2020) with designs reliable enough to detect an effect equal to half their obtained estimate and compute the exaggeration of significant estimates under this hypothesis. Since Ioannidis, Stanley and Doucouliagos (2017) finds typical exaggeration of two in economics, true effects may be, on average, half the magnitude of obtained estimates. I do not claim this is the true effect but rather assess power and exaggeration under this plausible assumption.[4] Under this assumption, the median power and exaggeration ratio in Brodeur, Cook and Heyes (2020) would be 37% and 1.6 respectively. Only 22% of designs would exceed the 80% power threshold, while 28% would exaggerate true effects by more than a factor of 2 on average.[5] While exaggeration is limited in some settings, it can be substantial in many others. Results are relatively comparable across methods (IV, DiD, RDD, and RCT).

Although the previous hypothesis regarding true effect sizes provides a broad overview of the literature, it may seem somewhat arbitrary. To reduce the burden of assumptions, I narrow the focus to IV designs, which enable a more systematic intra-study comparison between 2SLS estimates and the corresponding "naive" OLS ones. I do not claim that the OLS estimate accurately represents the true effect, but it may be reasonable to expect the IV to have enough power to detect an effect of the magnitude of the OLS point estimate. To explore this question, I leverage data from Young (2022). IV designs of significant headline results would have a median power of only 25% to detect an effect size of the magnitude of the OLS estimate. Significant estimates from half of these designs would exaggerate the OLS estimate by a factor larger than 2, and a quarter would do so by a factor larger than 3.5. Only 17% of these designs would reach the conventional 80% power threshold. These figures further underline heterogeneity in vulnerability to exaggeration and power issues. While many analyses are likely immune to exaggeration, a substantial share would not be powered enough to detect effects of realistic magnitude, leading significant estimates to considerably exaggerate these effects.[6] These results complement the analysis in Young (2022), which highlights the imprecision of published 2SLS estimates, making them statistically indistinguishable from the corresponding OLS estimates despite the fact that

4. This approach is also conservative to some extent: hypothesized effect sizes based on exaggerated estimates will be larger, thus minimizing estimated exaggeration.

5. Exaggeration is computed by drawing repeatedly from a normal distribution centered on the hypothetical effect size with standard deviation equal to the study's standard error, then computing the average of draws 1.96 standard errors from zero. I further discuss this approach and calculations in Section V. 2.2.

6. I find similar results in the literature on acute health effects of air pollution and in the political science literature, leveraging reviews implemented in Bagilet (2023) and Lal et al. (2024) respectively. The experimental literature displays less exaggeration on average but still substantial heterogeneity (Camerer et al. 2016). These analyses are described on the project's website.

2SLS point estimates substantially differ in magnitude from their OLS counterparts and are even regularly of the opposite sign.

II. 2 Illustration of the trade-off

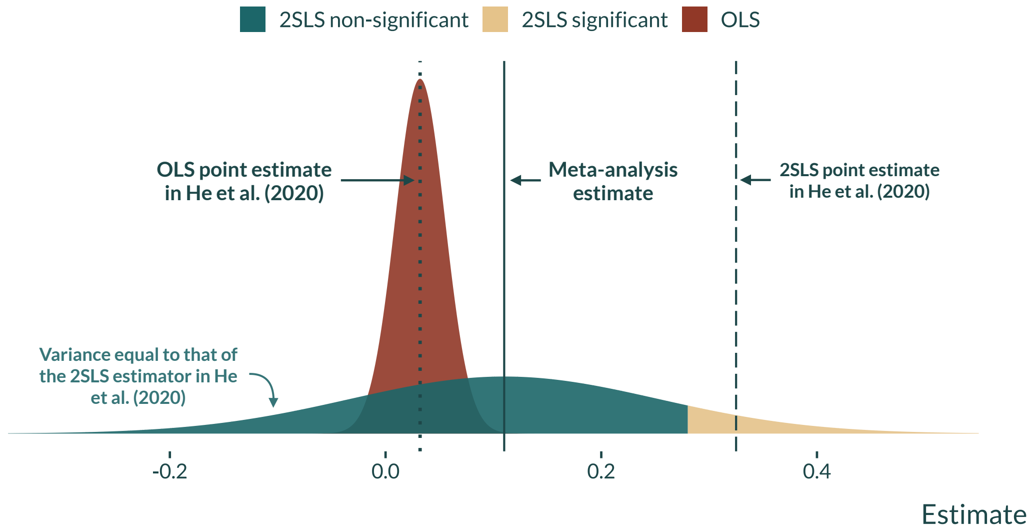
In order to limit hypotheses and to investigate causal exaggeration further, I focus on an example IV study investigating the impact of PM2.5 air pollution on mortality (He, Liu and Zhou 2020). It allows comparing the bias of the "naive" OLS to that of the IV by computing their distance to an estimate of the "true effect" they target. I define this "true effect" in two ways. First, I use the results of a meta-analysis of epidemiological studies (Shah et al. 2015). By pooling a number of studies carried out in various contexts, this meta-estimate might represent the average effect one may expect from such a study. However, the studies in this meta-analysis do not rely on canonical causal identification strategies used in economics and may be thought of as suffering from counfounding. I thus consider the result of Deryugina et al. (2019)—a precise causal study that may be less exposed to exaggeration—as an alternative estimate of the "true effect". This estimate may be context specific and the "true effect" in He, Liu and Zhou (2020) may deviate from these. The present discussion is conditional on this true underlying effect being close to these hypothesized true effect sizes.

He, Liu and Zhou (2020) finds that a "$10\mu g.m^{-3}$ increase in PM2.5 increases mortality by 3.25%" (s.e. 1.43%). Their corresponding OLS results suggest a 0.32% increase (s.e. 0.23%). For a similar increment in air pollution, Shah et al. (2015) and Deryugina et al. (2019) document a 1.1% and 1.8% increase in mortality respectively. The OLS estimate in He, Liu and Zhou (2020) is closer to the "true effect" based on Shah et al. (2015) than their 2SLS estimate. Provided that the three estimands are comparable, the bias of the IV is larger than that of the OLS. If the true effect was in fact closer to the one found by Deryugina et al. (2019), both biases would be roughly equal and the bias of the IV still substantial.

Exaggeration could explain this difference. Even if the 2SLS estimator effectively removes all conventional biases, the design in He, Liu and Zhou (2020) would still yield exaggerated statistically significant estimates. Figure 2 illustrates this point. Even if the 2SLS estimator is unbiased, i.e., centered on the meta-estimate found in Shah et al. (2015), the lack of precision of this design would lead significant estimates to substantially exaggerate the true effect, by a factor 3.2 on average. The 2SLS estimate found in He, Liu and Zhou (2020) could be one of these estimates.

This example illustrates that in a published study where a causal identification strategy substantially reduces the precision of the estimator, the resulting statistically significant estimates may be further away from the true effect than the "naive" OLS estimate, even if the estimator is

Figure 2 – Illustration of the Confounding-Exaggeration Trade-off in He, Liu and Zhou (2020)



Notes: The distribution for the 2SLS estimator is centered on the true effect, represented by the solid line and defined as the meta-estimate found in Shah et al. (2015). Its variance is equal to the one of the 2SLS estimator in He, Liu and Zhou (2020). The distribution for the OLS estimator is centered on the OLS estimate found in He, Liu and Zhou (2020) and its variance equal to that of this same estimator. The dashed and dotted lines represent the 2SLS and OLS estimates found in He, Liu and Zhou (2020) respectively. I ignore exaggeration of the OLS for clarity but since the OLS is biased downward, inflating OLS estimates bring them closer to the true effect.

unbiased. With this design, the average exaggeration would be 3.1 times larger than the OVB (or 1.3 time if the true effect is closer to the one found in Deryugina et al. (2019)).

III Mathematical derivation

In this section, I formally show the existence of the confounding-exaggeration trade-off and describe its determinants in a simple setting. I recall the definition of the exaggeration ratio and show that it increases with the variance of biased normally distributed estimators. I then compute the asymptotic distributions of a series of estimators—from a long and a short OLS regression, an IV and its reduced form—and study parameters driving their variances and thus their exaggeration. Finally, I show that, for any magnitude of OVB, there exist parameters and settings such that exaggeration is greater when using a causal inference method than the overall bias combining exaggeration and OVB in the naive regression.

III. 1 Properties of the exaggeration ratio

Following Gelman and Carlin (2014), one can define the exaggeration ratio E, as the expectation of the absolute value of significant estimates over the absolute value of the true effect. For an estimator $\hat{\beta}$ of a true effect β, with standard error σ and a two-sided hypothesis test of size α with threshold value z_α, let

$$E(\hat{\beta}, \sigma, \beta, z_\alpha) = \frac{\mathbb{E}\left[|\hat{\beta}| \, \middle| \beta, \sigma, |\hat{\beta}| > z_\alpha \sigma\right]}{|\beta|} \tag{1}$$

Lu, Qiu and Deng (2019) and Zwet and Cator (2021) showed that, for a given test and true effect sizes, the exaggeration ratio increases with the variance of an unbiased normally distributed estimator. This proof can easily be extended to biased estimators, in particular with additively separable bias.[7]

Lemma 1. *For an estimator $\hat{\beta}_b \sim \mathcal{N}(\beta + b, \sigma^2)$ of a true effect of magnitude β and a fixed additively separable bias b of the same sign as the true effect,*[8]

— *E is a decreasing function of the Signal-to-Noise Ratio (SNR) $\frac{\beta}{\sigma}$, i.e. the relative precision of the estimator, and depends on σ only through this SNR.*

— $\lim_{\sigma \to \infty} E(\hat{\beta}_b, \sigma, \beta, z_\alpha) = +\infty$.

In other words, the larger the variance of the estimator, the more exaggeration and, in the limit, exaggeration is infinite. Figure 1 provides a clear intuition for this result in the unbiased case. The goal of the present section is two-fold: 1) to describe drivers of exaggeration in causal settings, and 2) to show that the exaggeration bias of each causal design can be larger than any OVB in the corresponding naive regression. To study the former, given Lemma 1, one only needs to describe the evolution of the variances of these estimators with the parameters of interest: the IV strength in an IV setting, the number of exogenous shocks in a reduced form and the correlation between the explanatory variable of interest and the omitted variable of interest. Regarding the second objective of establishing the existence of the trade-off, one can show that, for any given level of OVB, there exist parameter values for which the exaggeration from a causal identification strategy exceeds that OVB. These comparative statics isolate the role of design parameters in driving exaggeration, independently of any effect on confounding. In applied settings, parameter values are implicitly determined by the setting and the choice of instruments or control sets, which explains why they can differ across designs.

7. All the proofs of the lemma and theorems are reported in appendix A.

8. I focus on cases where bias and true effect have the same sign as this represents the conservative scenario where exaggeration and OVB do not offset each other.

III. 2 Setting and data generating process

Consider a usual linear homoskedastic regression model with an omitted variable. For any individual $i \in \{1, ..., n\}$,

$$y_i = \beta_0 + \beta_1 x_i + \delta w_i + u_i \tag{2}$$

where y is the outcome, x the explanatory variable, w an unobserved omitted variable, u an unobserved error term. $(\beta_0, \beta_1, \delta) \in \mathbb{R}^3$ are unknown parameters. β_1 is the parameter of interest.

I assume homogeneous treatment effects and homoskedasticity, along with the usual OLS assumptions (*i.i.d.* observations, finite second moments, positive-definiteness of $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$—with $\mathrm{x}_i = (1, x_i)'$— and u_i conditional mean-zero and uncorrelated with x_i and w_i). Assume that w_i is unobserved, correlated with x_i and that $\delta \neq 0$. To simplify the derivations, I further assume that the unobserved variable is centered, *i.e.* $\mathbb{E}[w_i] = 0$. I also assume that the variance of the component of w_i that is orthogonal to x_i (denoted $w_i^{\perp x}$) does not vary with x_i, *i.e.*, $\mathrm{Var}(w_i^{\perp x}|x_i) = \mathrm{Var}(w_i^{\perp x})$. Consider the following data generating process for x_i:

$$x_i = \mu_x + \gamma w_i + \epsilon_i \tag{3}$$

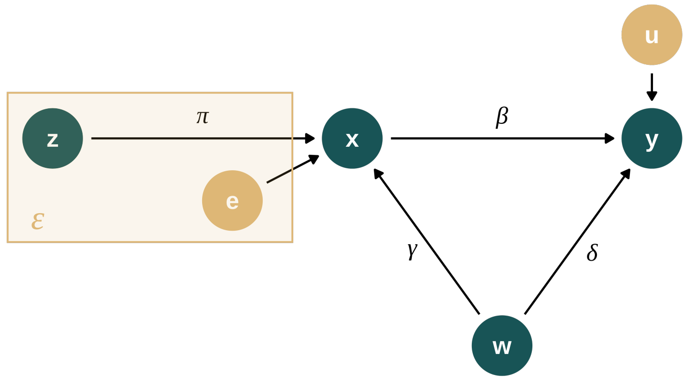
where $\gamma \in \mathbb{R}^*$ since x and w are correlated. Set $\rho_{xw} = \mathrm{corr}(x, w) = \frac{\gamma\sigma_w}{\sigma_x}$. In the IV and reduced form sections, I further assume that there exists a valid instrumental variable z_i for x_i, *i.e.* that $\mu_x + \epsilon_i = \pi_0 + \pi_1 z_i + e_i$ where $(\pi_0, \pi_1) \in \mathbb{R}^2$ are unknown parameters. The existence or not of this valid instrument does not affect the results in the controlled and OVB cases. Since the instrument is valid, it satisfies exogeneity, *i.e.* $\mathbb{E}[\mathrm{z}_i u_i] = 0$ and $\mathbb{E}[z_i w_i] = 0$, relevance, *i.e.* $\mathrm{rank}(\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']) = 2$, and positive-definiteness of $\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']$. The data generating process for x_i becomes:

$$x_i = \pi_0 + \pi_1 z_i + \gamma w_i + e_i \tag{4}$$

I assume that e_i is uncorrelated with z_i and w_i, *i.e.* $\mathbb{E}[z_i e_i] = 0$ and $\mathbb{E}[w_i e_i] = 0$. I also assume homoskedasticity for this term, such that $\mathbb{E}[e_i^2|z_i, w_i] = \sigma_e^2$ is constant.

Overall, this DGP is close to the usual textbook one but with an additional omitted variable. As such, it represents a simple baseline. Complexifying it, relaxing assumptions or accounting for aspects such as clustering or heteroskedasticity would typically increase estimator variance and thus exaggeration. The results presented here can therefore be viewed as lower bounds of the confounding-exaggeration trade-off. The Directed Acyclic Graph (DAG) in figure 3 represents the data generating process.

Figure 3 – DAG of the Data Generating Process



Notes: for clarity the error terms are represented in this graph, in beige. Model parameters are noted as edge labels.

III. 3 Asymptotic distributions of the estimators

For each model, the goal is to study the evolution of the sampling distribution variances with the value of the parameters of interest, for a fixed OVB. I assume that sample sizes are large enough so that the sampling distributions are well approximated by the asymptotic distributions. [9] To make comparisons across settings meaningful, I hold the variances of the "defined" variables ($\sigma_y^2, \sigma_x^2, \sigma_w^2$, and σ_z^2) fixed. When varying the parameters (γ, δ, or π_1), I adjust the error variances (σ_u^2 and σ_ϵ^2) so that the variances of the defined variables do not vary with the parameters. Only the proportions of their variance unexplained by the models do.

III. 3.1 Naive regression (OVB)

First, let us study the benchmark against which to compare the causal approaches and show that, for a given bias, its asymptotic variance does not depend on the parameters that will vary in other specifications (γ or π_1). Consider the "naive" regression of y on x (with w omitted).

Lemma 2. *Based on the data generating process described in Section III. 2, for $\hat{\beta}_{\text{OVB}}$ the OLS estimate of β_1 in the regression of y on x, $\sqrt{n}(\hat{\beta}_{\text{OVB}} - \beta_1) \xrightarrow{d} \mathcal{N}(b_{\text{OVB}},\ \sigma^2_{\text{OVB}})$, with*

$$b_{\text{OVB}} = \frac{\delta\gamma\sigma_w^2}{\sigma_x^2} \qquad \textit{and} \qquad \sigma^2_{\text{OVB}} = \frac{\sigma_u^2 + \delta^2\sigma_w^2(1-\rho_{xw}^2)}{\sigma_x^2}$$

The intuition for the formula of the asymptotic variance has been discussed in the introduction: $\sigma_u^2 + \delta^2\sigma_w(1-\rho_{xw}^2)$ is the part of the variance in y that is not explained by x ($\sigma^2_{y^{\perp x}}$).

9. In the IV case, convergence requires instruments not to be arbitrarily weak relative to the sample size. When instruments are very weak, finite-sample distributions may differ substantially from asymptotic distributions, typically exhibiting fatter tails. However, when power is low, significant estimates are located in the tails of the distribution and the additional mass in the tails leads to larger exaggeration.

Since σ_x^2 and σ_w^2 are fixed, reasoning at $b_{\text{OVB}} = cst$ is equivalent to considering that $\gamma\delta = \kappa = cst$. Noting that $\sigma_y^2 = \beta_1^2\sigma_x^2 + \delta^2\sigma_w^2 + 2\beta_1\kappa\sigma_w^2 + \sigma_u^2$, we can rewrite the variance of the estimator as

$$\sigma_{\text{OVB}}^2 = \frac{\sigma_y^2 - \beta_1^2\sigma_x^2 - 2\beta_1\kappa\sigma_w^2 - \kappa^2\frac{\sigma_w^4}{\sigma_x^2}}{\sigma_x^2}$$

For a fixed bias ($\kappa = cst$), σ_{OVB}^2 does not vary with γ and, based on Lemma 1, nor does E_{OVB}.

III. 3.2 Controlled regression

Next, let us turn to the "ideal" case in which no variable is omitted, *i.e.* we control for the omitted variable w and thus partial out confounders. This model accurately represents the DGP. This corresponds to the usual OLS setting with a constant and two regressors that are uncorrelated with the error: y regressed on x and w.

Lemma 3. *Based on the data generating process mentioned previously, for $\hat{\beta}_{\text{CTRL}}$ the OLS estimator of β_1 in the regression of y on x and w, $\sqrt{n}(\hat{\beta}_{\text{CTRL}} - \beta_1) \xrightarrow{d} \mathcal{N}(0,\ \sigma_{\text{CTRL}}^2)$, with*

$$\sigma_{\text{CTRL}}^2 = \frac{\sigma_u^2}{\sigma_x^2(1-\rho_{xw}^2)}$$

$\sigma_x^2(1-\rho_{xw}^2)$ is the part of the variance of x that is not explained by w ($\sigma_{x^{\perp w}}^2$) and σ_u^2 the part of the variance of y that is not explained by x nor w ($\sigma_{y^{\perp x,w}}^2$); here too we retrieved a result described in the introduction. For a given bias, we can rewrite σ_{CTRL}^2 as a function of fixed variances and one varying parameter, γ:

$$\sigma_{\text{CTRL}}^2 = \frac{\sigma_y^2 - \beta_1^2\sigma_x^2 - \frac{\kappa^2}{\gamma^2}\sigma_w^2 - 2\beta_1\kappa\sigma_w^2}{\sigma_x^2 - \gamma^2\sigma_w^2}$$

Since the numerator and denominator respectively increase and decrease with $|\gamma|$, σ_{CTRL}^2 increases with $|\gamma|$. For a given bias, the more w is correlated with x (and thus roughly the less it is with y since $\delta\gamma = const$), the larger the variance of the estimator. In addition, we can note that, for a given bias, the variance of the estimator can be arbitrarily large since $\lim_{\gamma^2 \to \frac{\sigma_x^2}{\sigma_w^2}} \sigma_{\text{CTRL}}^2 = +\infty$.

III. 3.3 Instrumental Variables

In the previous section, I considered a case in which we removed variation that included unwanted endogenous variation. Let us now turn to the IV, a converse situation where we select variation to keep, exogenous variation. Consider the IV model in which y is regressed on $\mathrm{x}_i = (1, x_i)'$ instrumented by $\mathrm{z}_i = (1, z_i)'$. This is a just-identified case and $\hat{\boldsymbol{\beta}}_{2\text{SLS}} = \hat{\boldsymbol{\beta}}_{\text{IV}}$.

Lemma 4. *Based on the data generating process mentioned above, for $\hat{\beta}_{\text{IV}}$ the IV estimator of β_1 in the regression of y on x instrumented by z, $\sqrt{n}(\hat{\beta}_{\text{IV}} - \beta_1) \xrightarrow{d} \mathcal{N}(0,\ \sigma^2_{\text{IV}})$ [10], with*

$$\sigma^2_{\text{IV}} = \frac{\sigma_u^2 + \delta^2\sigma_w^2}{\sigma_x^2\rho_{xz}^2}$$

The numerator is $\sigma^2_{y\perp\hat{x}}$, the part of the variance in y that is not explained by $\hat{x}$, the predicted value of x in the first stage and the denominator is $\sigma^2_{\hat{x}}$. For a given bias, noting that $\rho_{xz} = \text{corr}(x, z) = \pi_1 \frac{\sigma_z}{\sigma_x}$ and replacing σ_u^2, we can rewrite σ^2_{IV} as a function of fixed variances and one varying parameter, π_1:

$$\sigma^2_{\text{IV}} = \frac{\sigma_y^2 - \beta_1^2\sigma_x^2 - 2\beta_1\kappa\sigma_w^2}{\pi_1^2\sigma_z^2}$$

The smaller π_1, the larger σ^2_{IV}. In addition, $\lim_{\pi_1 \to 0} \sigma^2_{\text{IV}} = +\infty$.

III. 3.4 Reduced form

Let us now consider the regression of the outcome of interest directly on the instrument. Plugging equation 4 into equation 2 yields:

$$y_i = (\beta_0 + \beta_1\pi_0) + (\beta_1\pi_1)z_i + ((\delta + \beta_1\gamma)w_i + u_i + \beta_1 e_i)$$

Note that the resulting effect size is different from that of the other models. To make them comparable, one can set π_1 to 1 so that an increase of 1 in the instrument causes an increase of β_1 in y. Regardless of this assumption, regressing y on z corresponds to the usual bivariate, unbiased case and directly gives the following result:

Lemma 5. *Based on the data generating process mentioned previously, for $\hat{\beta}_{\text{RED}}$, the OLS estimator of the reduced form regression of y on z, $\sqrt{n}(\hat{\beta}_{\text{RED}} - \beta_1) \xrightarrow{d} \mathcal{N}(0,\ \sigma^2_{\text{RED}})$, with*

$$\sigma^2_{\text{RED}} = \frac{\sigma_y^2 - \beta_1^2\pi_1^2\sigma_z^2}{\sigma_z^2}$$

The numerator is the part of the variance of y that is not explained by z ($\sigma^2_{y^{\perp z}}$). In addition, the smaller σ_z^2, the larger σ^2_{RED} and $\lim_{\sigma_z \to 0} \sigma^2_{\text{RED}} = +\infty$.

In the binary case, $\sigma_z^2 = p_1(1 - p_1)$ with p_1 the proportion of treated observations, *i.e.*, the proportion of 1 in z. When most observations have the same treatment status, *i.e.*, p_1 close to 0

10. When instruments are arbitrarily weak however, the asymptotic distribution is non-normal, with fat tails. This type of distribution increases rather than decreases exaggeration, as underlined above.

or 1, σ_z^2 tends to zero and σ_{RED}^2 diverges. There is not enough variation in the treatment status to precisely identify the effect of interest.

III. 4 Exaggeration ratios

Combining the results from Lemma 2 through 5 regarding the asymptotic distribution of the various estimators with Lemma 1 stating that exaggeration increases with the variance of a normally distributed estimator yields:

Theorem 1. *For the data generating process described in Section III. 2, the exaggeration ratio of the controlled, IV and reduced form estimators, respectively $E_{\text{CTRL}}, E_{\text{IV}}$ and E_{RED}, are such that:*

- *E_{CTRL} increases with the correlation between the omitted variable and the explanatory variable of interest (i.e. $|\gamma|$ or $|\rho_{xw}|$), for a given bias,*
- *E_{IV} decreases with the strength of the IV (i.e. with $|\pi_1|$ or $|\rho_{xz}|$),*
- *E_{RED} increases when the number of exogenous shocks decreases in the binary case*

Each of these parameters determine the subset of variation in x leveraged for identification: the variation orthogonal to controls, that is explained by the instruments or that results from shocks. Each identification strategy restricts this variation, and the magnitude of restriction determines exaggeration. Using the same lemma and the limit properties of the variances described in Section III. 2, and since, at fixed bias, E_{OVB} does not vary with the parameters of interest, we get:

Theorem 2. *For the data generating process described in Section III. 2, $\forall\ b_{\text{OVB}}$,*

- *$\exists\ \gamma$ s.t. $E_{\text{CTRL}} > E_{\text{OVB}}$*
- *$\exists\ \pi_1$ s.t. $E_{\text{IV}} > E_{\text{OVB}}$*
- *$\exists\ \sigma_z$ s.t. $E_{\text{RED}} > E_{\text{OVB}}$*

In any setting, *i.e.* for any OVB, some causal designs can yield statistically significant estimates which are more biased than that of a naive regression. In other words and through the prism of identifying variation: causal identification strategies can remove so much variation that exaggeration becomes larger than the bias resulting from omitting any variable.

IV Simulations

To study the drivers of exaggeration in concrete settings, I build simulations that reproduce real-world examples from economics of education for RDD, political economy for IV, health economics for exogenous shocks and environmental and labor economics for control and fixed effects approaches. I split the simulations by identification strategy. Real-world settings enable clearly grasping the relationships between the different variables, setting realistic parameter values, based on existing studies and showing that exaggeration can arise for parameter values consistent with existing studies. To underline that causal exaggeration is not bound to specific studies, I do not reproduce a particular study but instead calibrate my simulations to emulate a typical non-experimental study from each literature. To check the representativity of my simulations, I compare their Signal-to-Noise Ratios (SNR) to the estimate/standard error ratios in studies from the corresponding literature. As underlined by Lemma 1, the SNR is a sufficient statistic for the exaggeration ratio; a SNR consistent with t-stats of the literature will ensure the representativity of the simulations with respect to the feature of interest, exaggeration. In these simulations, make many conservative simulation choices that ease the recovery of the effect of interest, limit estimation challenges and thus likely limit exaggeration. I consider simple linear models with constant and homogenous treatment effects, *i.i.d.* observations and homoskedastic errors. All the models are correctly specified and accurately represent the data generating process, except for the omitted variable, thus taking identification as given. Extensively documented code for each simulation procedure is available on the project's website.

IV. 1 Regression Discontinuity Design

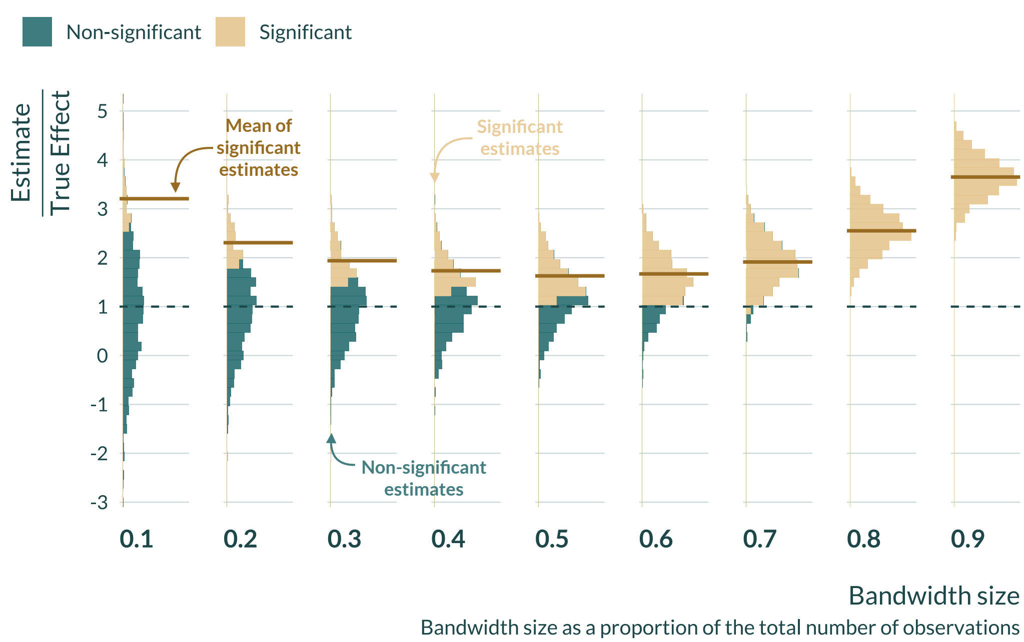
Intuition. A RDD relies on the assumption that for values close to the threshold, treatment assignment is quasi-random. It focuses on observations within a certain bandwidth around this threshold and discards observations further away. The effective sample used for causal identification is thus smaller than the total sample. A smaller bandwidth and effective sample size reduce precision and can create exaggeration. Here, the confounding-exaggeration trade-off is mediated by the size of the bandwidth.

Case-study and simulation procedure. To illustrate this trade-off, I consider a standard application of the sharp RD design from economics of education in which students are assigned to additional lessons based on the score they obtained on a standardized test. Thistlethwaite and Campbell (1960) introduced the concept of RDD using a similar type of quasi-experiment. Students with test scores below a given threshold receive the treatment while those above do not. Since students far above and far below the threshold may differ along unobserved characteristics such as ability, a RDD estimates the effect of the treatment by comparing outcomes of students whose initial test scores are immediately below and above this threshold.

The simulation framework is as follows. If a student i has an initial score $Qual_i$ below a cutoff C, they must take additional lessons, making the allocation of the treatment T sharp: $T_i = \mathbb{I}[Qual_i < C]$. Both qualification and final test scores are affected by students' unobserved ability w in a non-linear (cubic) way: a high or low ability has a strong positive impact on test scores while an average one does not strongly impact test scores. The qualifying test score of student i is thus: $Qual_i = \mu_q + \gamma f(w_i) + \epsilon_i$, where f a non-linear function (here cubic) and $\epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2)$ random noise. Their final test score is: $Final_i = \beta_0 + \beta_1 T_i + \eta Qual_i + \delta f(w_i) + u_i$, where β_1 is the causal parameter of interest. To ensure that ability impacts qualifying and final scores similarly, I set $\delta = \gamma(1 - \eta)$.

These simulations are built and calibrated to emulate a realistic study from this literature. I derive parameters of the distribution of grades from statistics from the Department of Education, treatment effect size is based on a meta-analysis of RCTs in economics of education (Kraft 2020). Sample and bandwidth sizes are consistent with an existing study leveraging an RDD to explore a similar question (Jacob and Lefgren 2004). I define ability such that it creates a large bias for large bandwidths and a limited bias for small bandwidths. Further details on calibration are available on the project's website. Given these parameters values, I generate 1000 datasets with 60,000 observations. For each dataset, I estimate the treatment effect by regressing the final score on the treatment status and the qualifying score for different bandwidth sizes.

Figure 4 – Evolution of the Bias with Bandwidth Size in the Regression Discontinuity Design Simulations, Conditional on Significance



Notes: 1000 iterations. Significativity level: 5%, N = 60,000. The brown lines represent the average of significant estimates. The bandwidth size is expressed as the proportion of the total number of observations in the entire sample. Details on the simulation are available at this link.

Results. Figure 4 displays the results of these simulations. For large bandwidth sizes, the distribution of estimates is far away from the true effect; there is an omitted variable bias. As the bandwidth size decreases, the identification strategy gets rid of the OVB and the distribution of the estimates centres on the true effet. At the same time, as the bandwidth and the effective sample size decrease, the distribution widens and significant estimates start representing a smaller and smaller subset of the distribution, located in the tails of the distribution, creating exaggeration. While the average of all estimates gets close to the true effect as bandwidth size and thus OVB decrease, in this setting the average of statistically significant estimates never gets close to the true effect. For large bandwidths, the omitted variable biases the effect while for small bandwidths, the small effective sample size creates exaggeration issues. The optimal bandwidth literature describes a similar trade-off but with different consequences (Imbens and Kalyanaraman 2012). They consider a bias-precision trade-off, I consider an omitted variable bias-exaggeration bias trade-off. Here, the parameter mediating the trade-off can directly be adjusted in a continuous way by researchers and the more we reduce one of these two biases, the more we increase the risk of facing the other. For a bandwidth size matching those found in the literature on which the simulations are calibrated, significant estimates would be 3.2 times larger than the true effect. [11]

IV. 2 Fixed Effects and Controlling for Confounders

Intuition. To identify a causal effect and avoid the risk of confounding, an "ideal" approach would be to partial these confounders out by directly controlling for them. However, as discussed in the introduction and Section III, controlling for an additional variable may increase the variance of the estimator if it absorbs more variation in the explanatory variable of interest than in the outcome variable. The same reasoning applies to a cornerstone causal identification approach: Fixed Effects (FEs). If including FEs partials out more of the variation in x than in y, it will increase the variance of the estimator and can lead to exaggeration.

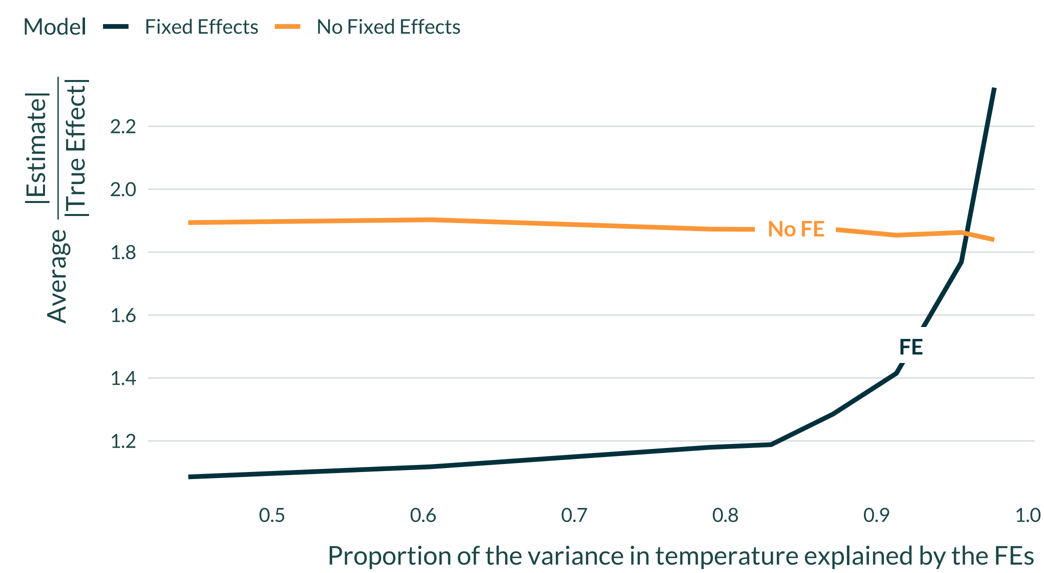
Case-study and simulation procedure. To highlight this trade-off, I consider the case of studies on the impact of temperature on worker productivity. A typical approach in this literature consists in estimating the link between different temperature bins and productivity, focusing in particular on high temperature bins (Lai et al. 2023, for a review of the literature). Usual approaches to explore this question typically rely on High Dimensional Fixed Effects models, including time and location fixed effects to adjust for invariant characteristics such as seasonal demand or location differences. For readability, in my simulations I only consider time fixed effects and abstract from

11. A typical article studying a similar question has an optimal bandwidth yielding roughly 5000 observations, which corresponds to a bandwidth of 0.1 in my simulations.

location ones. I assume that the true data generating process for temperature at time t in period τ is $Temp_{t\tau} = \mu_{Temp} + \gamma\lambda_\tau + \epsilon_t$ where μ_{Temp} is an intercept representing the average temperature and λ_τ are period fixed-effects with mean 0 and variance 1, such that their variance can be modified *via* γ. This particular structure of fixed effects allows controlling for the intensity of the fixed effects via the parameter γ, determining the proportion of the variation in temperature that comes from period-to-period (month) variations in average temperature. The productivity of worker i is defined as $Prod_{it\tau} = \beta_0 + \eta_i + \delta\lambda_\tau + \beta_1 \cdot \mathbb{1}\{Temp_{t\tau} \in (T_L, T_H]\} + u_{it}$ where η_i are individual fixed effects and $(T_L, T_H]$ the temperature bin of interest. The effects for other bins are not simulated and set to zero. For the omitted variable bias to remain constant when γ varies, I fix $\kappa = \gamma\delta$. I also adjust for the variances of ϵ and η to keep the variance of x and y constant.

I calibrate my simulations to replicate the distribution of the variables and interactions between variables from a typical study from the literature of interest (Stevens 2017, Somanathan et al. 2021, Lai et al. 2023, LoPalo 2023, among others). A more detailed description of calibration and modeling choices is available on the project's website. For each value of γ, or equivalently each proportion of the variation in temperature that is period-specific considered, I generate 1000 datasets with 49,000 observations and estimate two models, one including and the other omitting period fixed effects.

Figure 5 – Evolution of the Bias of Statistically Significant Estimates with the Proportion of the Variance in Temperature Explained by the Fixed Effects, in the FE Simulations



Notes: 1000 iterations. Significativity level: 5%, N = 49,000. The blue line indicates the average bias ratio for statistically significant estimates from the model including period FEs. The orange line represents the same quantity but for the model without the fixed effects. Details on the simulation and calibration are available at this link.

Results. Figure 5 displays the results of these simulations. When the (period) fixed effects explain a large part of the variation in temperature, the estimator is imprecise and leads to exaggeration to the point that it becomes larger than the simulated OVB. In actual settings, month fixed effects explain up to 90% of the variation in temperature, a value associated with an exaggeration factor of 1.3 in these simulations. However, the simulations do not model location fixed effects, which would likely further increase exaggeration. In addition, this setting features relatively large sample sizes; settings with a smaller one would likely exhibit substantial exaggeration even when fixed effects absorb a smaller share of variation.

IV. 3 Instrumental Variables Strategy

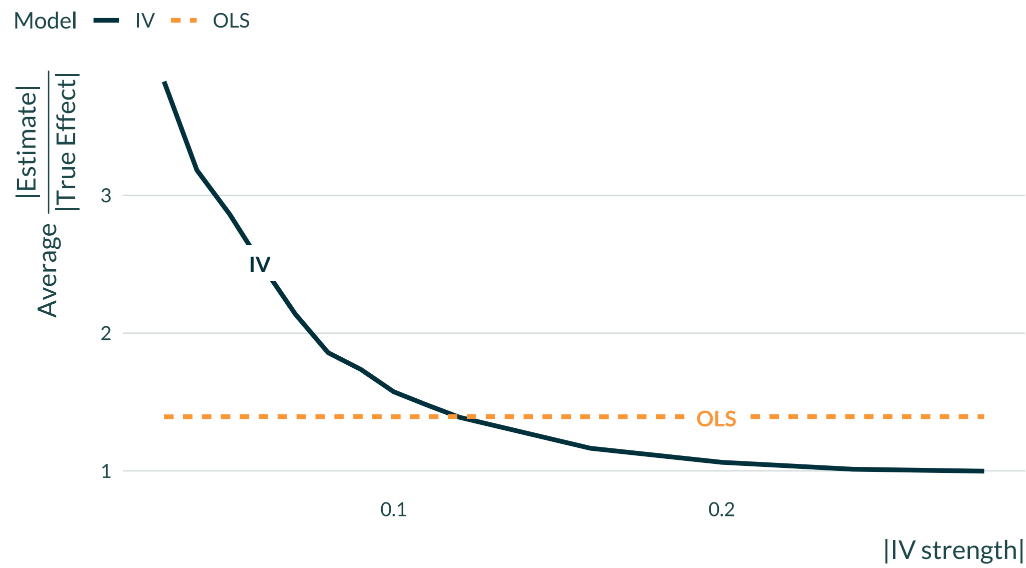
Intuition. Instrumental variables strategies overcome the issue of unobserved confounding by only considering the exogenous variation in the treatment, *i.e.* the variation that is explained by the instrument. Even when this exogenous fraction of the variation is limited, the instrument can successfully eliminate confounding on average. However, in such cases, the IV estimator will be imprecise and statistical power low. In the case of the IV, the confounding-exaggeration trade-off is mediated by the strength of the instrument considered. The weaker the instrument, the more inflated statistically significant estimates will be.

Case-study and simulation procedure. To illustrate this trade-off and its drivers, I consider the example of the impact of voter turnout on election results. To avoid the threat of confounding in this setting, existing studies take advantage of exogenous factors such as rainfall that affect voter turnout.[12] I reproduce such setting and assume that the true data generating process for the republican vote share is such that in location i, $Share_i = \beta_0 + \beta_1 Turnout_i + \delta w_i + u_i$, where w is an unobserved variable and $u \sim \mathcal{N}(0, \sigma_u^2)$ some random noise. The causal parameter of interest is β_1. In addition, turnout is affected by the amount of rain: $Turnout_i = \pi_0 + \pi_1 Rain_i + \gamma w_i + e_i$, where $Rain_i \sim \Gamma(\alpha, \theta)$ is the amount of rain in location i on the day of the election and e some random noise drawn form $\mathcal{N}(0, \sigma_e^2)$. I refer to π_1 as the strength of the instrumental variable.

To make the simulations realistic, I calibrate them on existing studies. I derive sample size, distribution parameters and effect sizes from a set of studies using similar variables (Gomez, Hansford and Krause 2007, Hansford and Gomez 2010, Fujiwara, Meng and Vogl 2016, Cooperman 2017). Details on the calibration choices are available on the project's website. For each value of IV strength considered, I create 1000 datasets of 30,000 observations. I run both a naive OLS and a 2SLS model to estimate the impact of voter turnout on republican vote share.

12. I abstract from potential exclusion restriction violations of this instrument and simulate it as exogenous.

Figure 6 – Evolution of the Bias of Statistically Significant Estimates Against Strength of the Instrument in the IV Simulations



Notes: 1000 iterations. Significativity level: 5%, N = 30,000. The blue line indicates the average bias ratio for IV statistically significant estimates. The orange line represents the same quantity for the OLS. Details on the simulation are available at this link.

Results. Figure 6 displays, for different IV strengths, the average of statistically significant estimates scaled by the true effect size for both the IV and the naive regression model. When the instrument is strong, the IV recovers the true effect, unlike the naive regression model. Yet, as IV strength decreases, the exaggeration of statistically significant estimates increases. Even when the omitted variable bias is large, exaggeration can exceed this bias for weak instruments. IV strengths similar to those observed in the literature on which the simulations are calibrated would yield an exaggeration factor of approximately 2.9. When the only available instrument is weak, using the naive regression model might, on average, produce statistically significant estimates closer to the true effect size than the IV. Importantly for applied research, a large first-stage F-statistic does not necessarily mitigate this problem. This result complements findings on the limitations of first-stage F-statistics with non-iid errors (Young 2022, Lal et al. 2024).

IV. 4 Exogenous shocks

Intuition. Taking advantage of exogenous variation in treatment status induced by shocks or events can also help avoid confounding. In many settings, while the number of observations may be large, the number of events, their duration or the proportion of individuals affected might be limited. As a consequence, the number of (un)treated observations can be small and the variation available to identify the treatment limited. As extensively discussed in the randomized controlled trial literature, statistical power is maximized when the proportion of treated observations is equal to the proportion of untreated ones and drops when one of these proportions gets close to 0. In studies using discrete exogenous shocks, a confounding-exaggeration trade-off is thus mediated by the number of treated observations.

Case-study and simulation procedure. To illustrate this trade-off, I simulate a study of the impact of air pollution reduction on newborn weight of babies. To avoid confounding, one can exploit exogenous shocks to air pollution such as plant closures, creation of a low emission zone or of an urban toll (Currie et al. 2015, Lavaine and Neidell 2017). I simulate a typical analysis, at the zip code and monthly levels and focus on the example of toxic plant closures. I consider that the average birth weight in zip code z at time period t, bw_{zt}, depends on a zip code fixed effect ζ_z, a time fixed effect τ_t, and the treatment status T_{zt}, equal to one if a plant is closed in this period and 0 otherwise. The average birth weight in zip code z at time t is defined as follows: $bw_{zt} = \beta_0 + \beta_1 T_{zt} + \zeta_z + \tau_t + u_{zt}$. To further facilitate the identification of the effect, I assume a random treatment allocation across periods and zip codes as well as constant and homogenous effects. I only vary the proportion of observations affected by toxic plant closings.

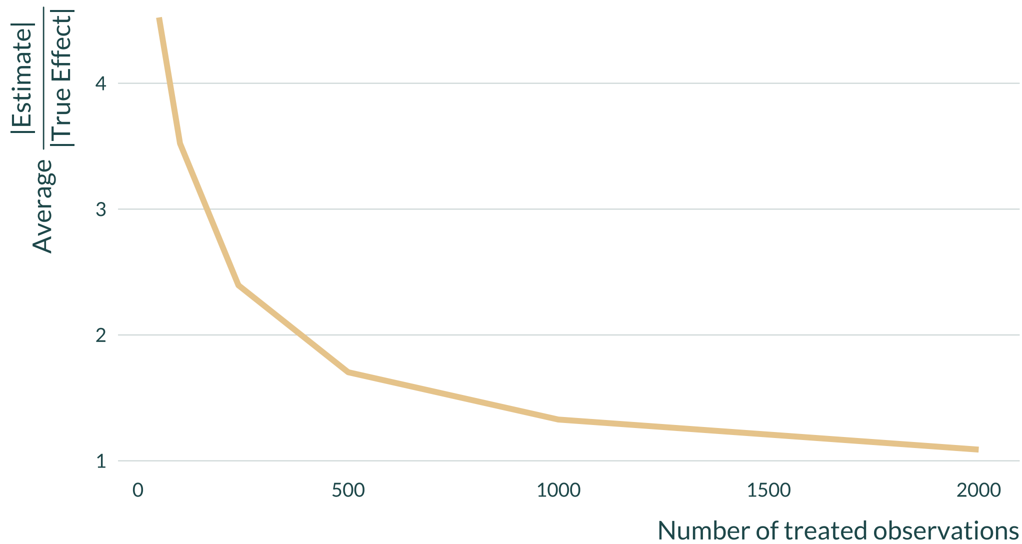
I derive parameters values from studies from the literature (Currie et al. 2015, Lavaine and Neidell 2017, Yang and Chou 2018). The simulations mimic the design of such studies, considering a similar number of observations (100,000), distributions of variables, treatment allocation procedure and treatment effect sizes. I generate datasets with an increasing number of treated observations, varying the proportion of treated units and estimate the correct two-way fixed effects model. The simulation procedure is described in details on the project's website.

Results. Figure 7 displays the results of these simulations. As expected, exaggeration is larger for a smaller number of treated observations, keeping every thing else constant. Even though the actual sample size is extremely large in this example, if the number of treated observations is small, as it can be the case in this literature, exaggeration can be substantial. For instance, one of the papers in this literature displays 240 treated observations (out of 120,000), a scale that would produce in my analogous, yet conservative, simulated setting an exaggeration ratio of about 2.4. A very large number of observations does not necessarily shield us from exaggeration.

V Navigating the trade-off

In the previous sections, I showed that using causal identification strategies induces a trade-off between avoiding confounding and exaggerating true effects. How can we, as applied researchers using observational data, arbitrate it? Since key pieces of information such as the true effect and

Figure 7 – Evolution of Bias With the Number of Treated Observations, for Statistically Significant Estimates, in the Exogenous Shocks Simulations



Notes: 1000 iterations. Significativity level: 5%, N = 100,000. The line indicates the average bias ratio of statistically significant estimates. Details on the simulation are available at this link.

the effect of omitted variables are inherently unknown, we cannot directly compute the biases caused by confounders and exaggeration. In this section, I examine how we can, however, get a sense of threats from both sides of the trade-off and probe its main driver, the variation used for identification. I then discuss how changing attitudes towards statistical significance and replicating studies could limit the exaggeration issue.

V. 1 Gauging omitted variable bias

On one side of the trade-off lies the widely discussed bias caused by confounders. Although it is in essence impossible to measure, tools such as sensitivity analyses are available to gauge its magnitude (Rosenbaum 2002, Middleton et al. 2016, Oster 2019, Cinelli and Hazlett 2020). For instance, the method developed in Cinelli and Hazlett (2020) enables assessing how strong confounders would have to be to change the estimate of the treatment effect beyond a given level we are interested in. A typical conclusion from such an analysis would be: "omitted variables would have to explain as much residual variance of the outcome and the treatment as the observed covariate x (age for instance) to bring down the estimate to a value of β_l". The authors also implement graphical tools to facilitate this comparison. I suggest using quantitative bias analyses to assess how restrictive a causal approach must be to limit unobserved confounding to acceptable levels. Extremely strict controls and fixed effects may leave little scope for confounders but can generate substantial exaggeration. In such situations, the potential influence of confounders should be weighed carefully: given the risks of exaggeration—and concerns about representativity—causal estimates should not be treated as definitive benchmarks for assessing naive estimates, nor should a discrepancy between the two be taken as sufficient evidence that the naive estimate is incorrect.

V. 2 Evaluating risks of low power and exaggeration

On the other side of the trade-off lies exaggeration, which emerges when statistical power is low. Like OVB, exaggeration and statistical power are in essence impossible to measure as their computation depends on the true effect, which is always unknown. Yet, power calculations can help assess them by making assumptions about the magnitude of the true effect. In randomized controlled trials, such computations are not only an established practice but a requirement (Duflo, Glennerster and Kremer 2007, McConnell and Vera-Hernández 2025, Athey and Imbens 2017). They are, however, rarely reported in non-experimental studies. In causal inference textbooks, very few pages are devoted to statistical power in non-experimental studies (Angrist and Pischke 2009; 2014, Imbens and Rubin 2015, Cunningham 2021). To the best of my knowledge, only two textbooks discuss the matter in depth (Shadish, Cook and Campbell 2002, Huntington-Klein 2021). Yet, taking publication bias and the threat of exaggeration into account highlights the necessity of running power calculations in non-experimental studies as well. Low power or a large variance not only makes it more difficult to detect an effect or to draw clear conclusions about its magnitude when detected but it can also create a bias. To avoid this bias, I advocate making power more central to non-experimental analyses, even after a significant estimate has been obtained. Results from power and exaggeration calculations would be both highly informative and straightforward to report in the robustness sections of papers. I next discuss how to run such calculations.

V. 2.1 Prospective power calculations

To evaluate the statistical power of a study, the risk of exaggeration, and the factors driving it, one can simulate the design of the study before implementing it (Hill 2011, Gelman 2020, Black et al. 2022). Simulating a data generating process from scratch requires thinking about the distribution of the variables and their relationships, which can help identify the variation used for identification and reveal potential limitations of the design. I implemented such Monte Carlo simulations in Section IV. The replication material and `R` code I provide can be used as templates to implement simulations for most causal identification strategies, based on data generated from scratch. In situations where the relationships among covariates are too complex to emulate, one can start from an existing dataset and add a fake treatment effect to the data. I implemented such real-data simulations in a companion paper and describe their implementation in its replication material (Bagilet 2023).

V. 2.2 Retrospective power calculations

Post-analysis power calculations can also help assess the statistical power of a research design, as illustrated in Section II. Such retrospective calculations allow evaluating whether the design would produce accurate and uninflated statistically significant estimates if the true effect were smaller than the observed estimate (Gelman and Carlin 2014, Ioannidis, Stanley and Doucouliagos 2017, Stommes, Aronow and Sävje 2023).

I illustrate this approach using Card (1993), who finds that an additional year of education causes a 13.2% increase in wage (s.e. 5.5%), instrumenting education with distance to a four-year college during childhood. As the author notes, the estimate is imprecise. Suppose the existing literature suggests that such effects are typically closer to 10%. We can use power calculations to assess whether the design in Card (1993) would detect an effect of this magnitude.[13] Under this 10% hypothesis, the average of statistically significant estimates at the 5% level would be approximately 15%, overestimating the true effect by a factor of 1.5. Statistical power would be only 44%. Under this assumption, the design would be underpowered and exaggeration substantial.

The usefulness of retrospective power analysis depends on assumptions about the true effect size. Plausible effect sizes can be derived from meta-analyses, credible studies (*e.g.*, large RCTs), or theoretical findings. When meta-analyses are available, Bayesian shrinkage can adjust estimates based on prior evidence (Zwet and Gelman 2021, Zwet and Cator 2021, Zwet, Schwab and Senn 2021). When such information is not available, we can evaluate power for a range of smaller but credible effect sizes. When defining such alternative effect sizes, we should keep in mind that any obtained estimate or previously published ones may themselves be exaggerated (Gelman and Carlin 2014).

V. 2.3 Precision matters even after obtaining a significant estimate

In non-experimental studies, estimator variance often plays a critical role because a large variance may prevent rejecting the null hypothesis when it is false. As a result, variance is typically a concern but only until a statistically significant estimate is obtained. However, exaggeration highlights that variance remains important for the reliability of the point estimate—and for reasons beyond mere uncertainty—even after statistical significance has been achieved.

13. Timm, Gelman and Carlin (2019) and Linden (2019) offer `R` and `Stata` packages that enable easily running these calculations through an extremely short command: `retrodesign(10, 5.5)`. Exaggeration is computed by drawing repeatedly from a normal distribution centered on the hypothetical effect size of 10% with a standard deviation equal to the standard error of the estimate found in the study (here 5.5%), then computing the average of the absolute value of the draws that are 1.96 standard errors away from zero.

Obtaining a statistically significant estimate from an imprecise estimator should not necessarily be interpreted as a sign of "success" in getting significance despite a wide confidence interval. Rather, it may signal that this estimate lies in the tails of the distribution and thus inaccurately represents the true effect. This observation reframes the conventional bias-variance trade-off as a bias-bias trade-off: a larger variance can lead to a larger bias, even in (conditional) expectation. This paper therefore calls for careful attention to the implications of design and modeling choices on estimator variance, even when large variance has not prevented statistical significance.

V. 3 Identifying the identifying variation

V. 3.1 A metric for identifying observations

The approaches outlined above for assessing exaggeration and confoundings are built on assumptions regarding the true effect size. To circumvents these assumptions, one can instead directly focus on the driver of the bias-variance trade-off: the variation used for identification. When the identifying variation is too limited, statistical power is low and exaggeration is large. Pinpointing the specific variation on which estimation relies can provide a direct way to gauge the risk of exaggeration. This section introduces tools to identify that variation.

To identify the identifying variation in an applied study, one usually begins with a thought exercise: where does the variation leveraged for identification come from? This reflection is integral to the implementation of any applied economics study. While it is typically undertaken primarily for identification and internal validity purposes, this paper highlights its importance for exaggeration reasons, *i.e.* after a significant estimate has been obtained. Although this exercise may be straightforward in simple cases–such as one-way fixed effects–it becomes more challenging in complex designs involving multiple or interacting fixed effects, especially for readers less familiar with the setting.

Given this difficulty of intuitively identifying the source of variation in complex models, it can be useful to develop measures that quantify the contribution of each observation or group of observations to the estimation of the treatment effect. A range of existing tools from the statistics literature, such as leverage and Cook's distance, measure the influence of individual observations on regression parameters. These measures, however, assess influence on the *entire* parameter vector and are not directly suited for applied economics where interest is typically confined to a single parameter: the coefficient of the treatment variable. To get to a more suited measure, I propose to first apply the Frisch-Waugh-Lovell theorem and then compute leverage for the regression of the residualized outcome on the residualized treatment. The residuals are obtained from regressions on the full set of controls, including fixed effects and other identification-related controls such as control functions. This produces observation-specific weights describing the extent to which each observation contributes to the estimation of the treatment effect.

These identifying-variation weights are equivalent–up to a normalization to one–to the multiple regression weights introduced in Aronow and Samii (2016) and previously discussed in Angrist and Pischke (2009). The estimate of the treatment coefficient in a simple linear regression can be interpreted as a weighted average of individual treatment effects. The weight ω_i of individual i is the squared difference between its treatment status T_i and the value of this treatment status as predicted by the other covariates X: $\omega_i = (T_i - \mathbb{E}[T_i|X_i])^2$. At the group level, these weights connect to well-established insights, particularly in the widely used case of fixed effects. A group's weight, *i.e.* the sum of the individual weights within that group, corresponds to the within-group variance of the conditional treatment status. Since including fixed effects changes the type of variation used for identification from pooled to within-group variation, groups with little within variation in treatment status only marginally contribute to identification. These weights also have crucial implications at the observation level: individual observations whose treatment status is largely explained by covariates contribute little, if at all, to estimation. Variation in the explanatory variable of interest is absorbed by the controls. In that sense, the weights not only highlight the role of groups but also the uneven contribution of individual observations within them. More broadly, in settings without a natural grouping structure, observation-level weights help assess how much each unit actually contributes to identification. Aronow and Samii (2016) uses the multiple regression weights to argue that, when treatment effects are heterogeneous, some observations may be disproportionately represented in the estimated average effect, raising concerns about external validity. Even in the absence of heterogeneity, however, relying on observations whose treatment status is well explained by covariates can yield a very small *effective* sample size and, in turn, exaggeration. Whereas Aronow and Samii (2016) emphasizes the representativity of the effective sample for external validity reasons, the present paper focuses on its size, for fear of exaggeration. It also does not focus on high-weight observations but instead on the lack of contribution of low-weight ones.

V. 3.2 A tool to identify the identifying variation

Although the computation of these weights is relatively straightforward, it involves several steps, and their subsequent analysis and comparison are more challenging. To facilitate this process, I develop the `ididvar` package for `R`.[14] It provides a unified tool to compute identifying-variation weights and related contribution metrics, along with a series of visualization tools. By lowering the cost of identifying which observations drive estimation, the package aims helping assess potential

14. More information on the package are available on the package's website. This package can be extended to accommodate more specific settings. Readers are encouraged to suggest improvements through GitHub issues or pull requests.

risks of exaggeration when identifying variation is limited. Unlike power calculations, these metrics can be computed without assumptions about the true effect size.

Most functions are high-level and easy to use, requiring only the regression output and the variable of interest. A first set of functions computes and visualizes weights, including their distribution across space, groups, or time. This enables users to detect low-weight observations or clusters that contribute little to identification. Because the distribution of weights is often heterogenous, building meaningful visualization can be challenging. To address this, I propose a discretized and logged scale that compares each weight to the average weight, $1/n$, where n is the number of observations or groups. This representation clarifies visualization and ensures comparability across specifications and studies.

To illustrate this feature of the package, I consider a simple example using a country-level panel (Bryan 2025, restricted to Africa and Europe for clarity) and estimate the within-country relationship between GDP per capita and life expectancy: $\log(PerCapitaGDP_{ct}) = \alpha_c + \beta LifeExp_{ct} + u_{ct}$. Figure 8 presents visualizations produced by the package. Panel A displays the identifying-variation weights for each observation, [15] while Panel B shows their cumulative distribution. The distribution is highly heterogeneous: many observations contribute little to identification, particularly those from mid-periods and European countries.

Weights alone, however, do not indicate which observations actually contribute to identification. A second set of functions identifies the *effective sample*: the smallest subset of observations that cannot be removed without altering the point estimate or standard error of the parameter of interest by more than a chosen margin, e.g., 5%. The procedure iteratively re-estimates the model, removing low-weight observations following an approach similar to influence-function methods. Reporting the effective sample size helps interpret results in light of the sample that actually drives identification, much as one would interpret the nominal sample size.

In the illustrative example, Panel C of Figure 8 shows how point estimates and standard errors evolve as low-weight observations are sequentially dropped. 70% of the nominal sample can be dropped without affecting estimates and standard errors by more than 5%. The effective sample is thus about three times smaller than the nominal sample ($n_{\text{nominal}} = 984$ and $n_{\text{effective}} = 246$). Results should be interpreted accordingly. Panels D and E characterize this effective sample: many observations from Africa and mid-periods do not contribute to identification, implying that the effective sample, in addition to be small, is not representative of the full dataset.

The package is flexible and can be applied to a wide range of identification strategies and estimation methods. It enables users to compare specifications and explore how the inclusion of

15. This graph is essentially generated in one line of code: `idid_viz_weights(reg, ''l_gdpPercap'', year, country)` where `reg` is the output of the regression of interest, `l_gdpPercap` the variable of interest and `var_x` and `var_y` specify the variables to be displayed n the x and y-axes respectively

different sets of fixed effects or controls affects the effective sample. More broadly, by making the structure and amount of identifying variation explicit, the package provides a concrete way to assess where a study lies along the confounding-exaggeration trade-off. In doing so, it aims to help users evaluate whether the credibility gained from additional controls comes at the cost of a loss of identifying variation and statistical power.

V. 4 Attitude Towards Significance and Replication

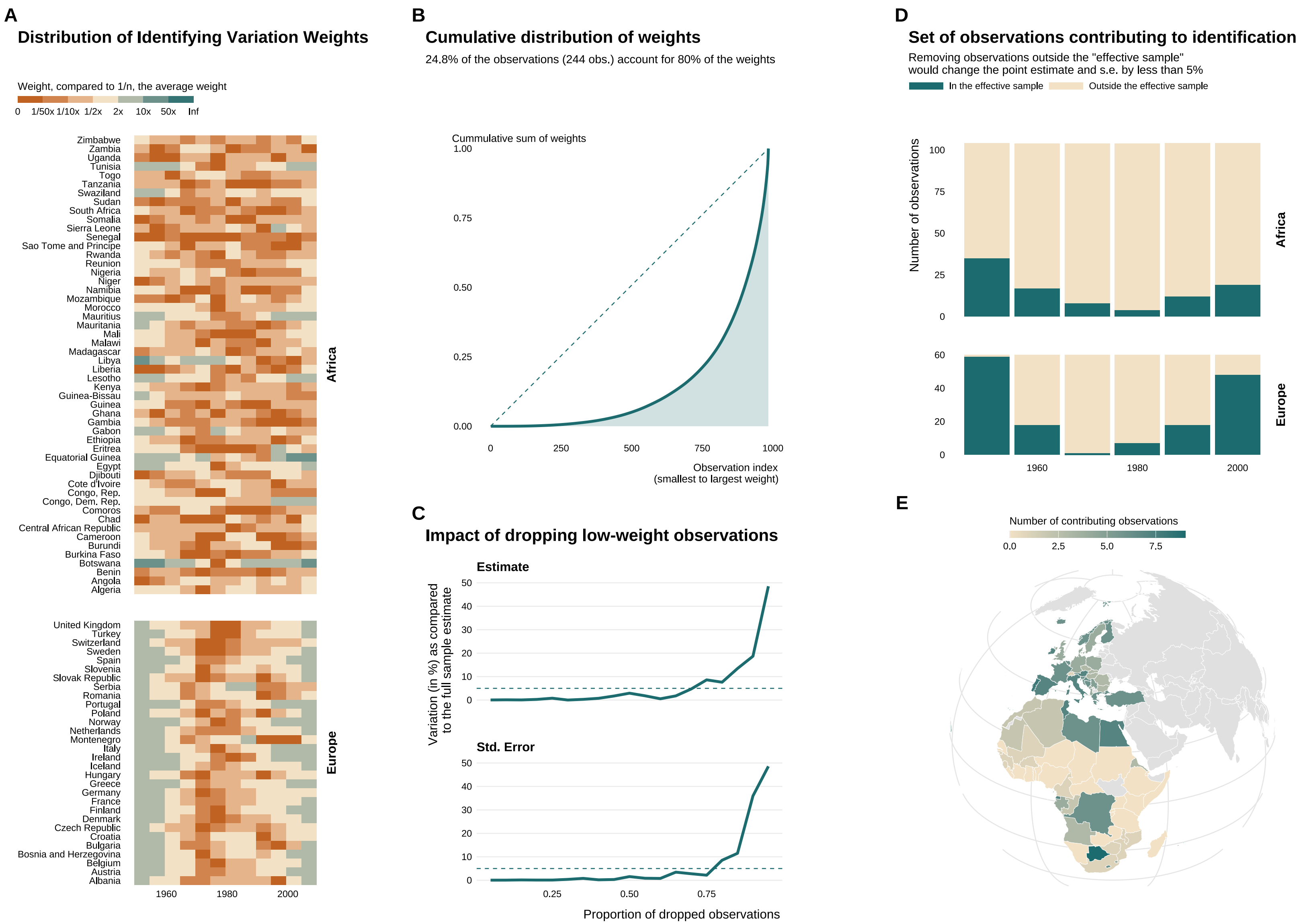
Exaggeration only arises in the presence of publication bias. As shown in the simulations, if estimates were not filtered by their statistical significance nor any other characteristics, even underpowered studies would on average recover the true effect, as long as the estimator is unbiased. The exaggeration issue could therefore be addressed by tackling publication bias. [16]

To identify broader pathways to eliminate this filtering of significant results, it is helpful to first understand how statistically significant results arise when power is low. In such situations, they can be obtained either by chance or as an outcome of the garden of forking paths (Simmons, Nelson and Simonsohn 2011, Gelman and Loken 2013, Kasy 2021). Decision forks appear at various stages along the path of research: for instance, in data preparation, when deciding whether to include a control variable or observation in the sample, or when deciding whether to pursue research that yields non-significant results. Due to structural incentives that favor significance, the path followed may be more likely to lead to a statistically significant result. These choices most often do not reflect poor researcher practices but rather emerge from a system that treats significant results as an end goal of research.

Since the issue is partly structural, system-level changes in scientific practices could alleviate exaggeration and the trade-off described in this paper. First, many researchers advocate not using statistical significance as a measure of study quality (McShane et al. 2019, Abadie 2020). To be effective, this change should be paired with efforts to replicate studies (Christensen and Miguel 2018). Replications, even of underpowered studies, would eventually enable constructing the distribution of the causal estimand of interest. Meta-analyses would then reduce uncertainty around the true value of the estimand by pooling estimates (Hernán 2022). Finally, inflation of statistically significant estimates could be limited by interpreting confidence intervals and not point estimates, thus considering these intervals as compatibility intervals, with their width indicating a range of effect sizes compatible with the data (Shadish, Cook and Campbell 2002, Amrhein, Trafimow and Greenland 2019, Romer 2020).

16. Note that even if this publication bias is often understood as selection of statistical significance, it can also extend to any situation in which a certain type of results, large or surprising for instance, are favored. The present paper focuses on the former type of selection owing to its primacy and the fact that its existence and the mechanisms causing it are well-documented.

Figure 8 – Illustrative Visualizations and Analyses Produced with `ididvar`



Notes: Details on the computation are available at this link.

VI Conclusion

The economic literature suffers from an extensive lack of statistical power (Ioannidis, Stanley and Doucouliagos 2017) and strongly favors statistically significant findings (Rosenthal 1979, Andrews and Kasy 2019, Vivalt 2019, Abadie 2020, Brodeur, Cook and Heyes 2020, for instance). In such situations, significant estimates from underpowered studies always exaggerate true effect sizes, even when the estimators are "unbiased" in the usual sense of $\mathbb{E}[\hat{\beta}] = \beta$ (Gelman and Tuerlinckx 2000, Ioannidis 2008, Gelman and Carlin 2014). It is therefore not surprising that many estimates published in economics have been found to be considerably exaggerated (Ioannidis, Stanley and Doucouliagos 2017), despite the extensive use of convincing causal inference methods. Yet, the determinants of these exaggeration and power problems have remained largely understudied. I argue that exaggeration is exacerbated by the foundational component of causal inference: the fact that it only leverages subsets of the variation for identification. Although causal methods enable avoiding confounding, they also reduce statistical power and thus increase the risk of exaggeration. The same aspect that makes these methods credible can create another type of bias. Systematically reporting statistical power and examining the variation actually used for identification could help avoid falling into this exaggeration trap.

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A Mathematical proofs

I. 1 Variation of the exaggeration ratio (Lemma 1)

Proof. Lu, Qiu and Deng (2019) and Zwet and Cator (2021) showed this in the case of $b = 0$. To extend it to the biased case, consider $E_b = \frac{\mathbb{E}\left[|\hat{\beta}_b| \;\middle|\beta_1, \sigma, |\hat{\beta}_b| > z_\alpha \sigma\right]}{|\beta_1|}$ the exaggeration ratio of interest. Note that, since $\hat{\beta}_b$ is an unbiased estimator of $\beta_1 + b$, $\tilde{E}_b = \frac{\mathbb{E}\left[|\hat{\beta}_b| \;\middle|\beta_1, \sigma, |\hat{\beta}_b| > z_\alpha \sigma\right]}{|\beta_{1+b}|}$ has the properties described in the lemma. Now, considering that $E_b = \left|\frac{\beta_1 + b}{\beta_1}\right| \tilde{E}_b$ shows the properties when β_1 and b have the same sign. ☐

I. 2 Asymptotic distribution of $\hat{\beta}_{\text{OVB}}$ (Lemma 2)

For readability, let us introduce the usual vector notation such that for instance $y = (y_1, ..., y_n)'$ and set $\boldsymbol{\beta} = (\beta_0, \beta_1)'$ and $\mathrm{x}_i = (1, x_i)'$. I also use capital letters to denote matrices (for instance $X = (\mathrm{x}_1', ..., \mathrm{x}_n')'$).

Proof. Since, we do not observe w, we consider the projection of y on X only:

$$y = X\boldsymbol{\beta}_{\text{OVB}} + u_{\text{OVB}} \tag{5}$$

where by definition of the projection, $\mathbb{E}[X'u_{\text{OVB}}] = 0$.

We first compute the bias of the estimator. From equation 5 we get:

$$\begin{aligned} & X'y = X'X\boldsymbol{\beta}_{\text{OVB}} + X'u_{\text{OVB}} \\ \Rightarrow\quad & \mathbb{E}[X'y] = \underbrace{\mathbb{E}[X'X]}_{\text{pos. def.}}\boldsymbol{\beta}_{\text{OVB}} + \underbrace{\mathbb{E}[X'u_{\text{OVB}}]}_{0} \\ \Leftrightarrow\quad & \boldsymbol{\beta}_{\text{OVB}} = \mathbb{E}[X'X]^{-1}\mathbb{E}[X'(X\boldsymbol{\beta} + \delta w + u)] \quad \text{cf eq. 2} \\ \Leftrightarrow\quad & \boldsymbol{\beta}_{\text{OVB}} = \boldsymbol{\beta} + \mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]\delta \end{aligned} \tag{6}$$

We then compute the asymptotic distribution. We can write:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{OVB}} - \boldsymbol{\beta}_{\text{OVB}}) = \left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i\mathrm{x}_i'\right)^{-1} \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i u_{\text{OVB},i}\right)$$

Applying the Weak Law of Large Numbers (WLLN), the Central Limit Theorem (CLT) and Slutsky's theorem yields:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{OVB}} - \boldsymbol{\beta}_{\text{OVB}}) \xrightarrow{d} \mathcal{N}\left(0, \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' u_{\text{OVB},i}^2]\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\right) \tag{7}$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{OVB}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}$, $\mathbb{E}[\mathrm{x}_i w_i]$ and $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' u_{\text{OVB},i}^2]$.

$$\mathbb{E}[\mathrm{x}_i\mathrm{x}_i'] = \mathbb{E}\begin{bmatrix} 1 & x_i \\ x_i & x_i^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_x \\ \mu_x & \sigma_x^2 + \mu_x^2 \end{bmatrix} \quad \Rightarrow \quad \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1} = \frac{1}{\sigma_x^2}\begin{bmatrix} \sigma_x^2 + \mu_x^2 & -\mu_x \\ -\mu_x & 1 \end{bmatrix}$$

$$\mathbb{E}[\mathrm{x}_i w_i] = \mathbb{E}\begin{bmatrix} w_i \\ x_i w_i \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbb{E}[x_i]\underbrace{\mathbb{E}[w_i]}_{0} + \text{cov}(x_i, w_i) \end{bmatrix} = \begin{bmatrix} 0 \\ \gamma\underbrace{\text{var}(w_i)}_{\sigma_w^2} + \underbrace{\text{cov}(\epsilon_i, w_i)}_{0} \end{bmatrix} = \begin{bmatrix} 0 \\ \gamma\sigma_w^2 \end{bmatrix}$$

$$\Rightarrow \quad \mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i] = \frac{\gamma\sigma_w^2}{\sigma_x^2}\begin{bmatrix} -\mu_x \\ 1 \end{bmatrix} \tag{8}$$

Note that $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' u_{\text{OVB},i}^2] \overset{\text{LIE}}{=} \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\mathbb{E}[u_{\text{OVB},i}^2|\mathrm{x}_i]]$. We thus first compute $\mathbb{E}[u_{\text{OVB},i}^2|\mathrm{x}_i]$, noting:

$$\begin{aligned}
u_{\text{OVB},i} &= y_i - \mathrm{x}_i'\boldsymbol{\beta}_{\text{OVB}} \\
&= \delta w_i + u_i + \mathrm{x}_i'(\boldsymbol{\beta} - \boldsymbol{\beta}_{\text{OVB}}) \\
&= \delta w_i + u_i - \underbrace{\mathrm{x}_i'\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i]}_{\text{projection of } w_i \text{ on } x_i}\delta \\
&= u_i + \delta\underbrace{\left(w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\right)}_{\text{part of } w_i \text{ orthogonal to } x_i} \\
&= u_i + \delta w_i^\perp \qquad\qquad \text{where } w_i^\perp = w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)
\end{aligned}$$

And thus,

$$\begin{aligned}
\mathbb{E}[u_{\text{OVB},i}^2|\mathrm{x}_i] &= \mathbb{E}[(u_i + \delta w_i^\perp)^2|x_i] \\
&= \mathbb{E}[u_i^2|x_i] + 2\delta\mathbb{E}[u_i w_i^\perp|x_i] + \delta^2\mathbb{E}[(w_i^\perp)^2|x_i] \\
&= \sigma_u^2 + 2\delta(\mathbb{E}[u_i w_i|x_i] - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\underbrace{\mathbb{E}[u_i|x_i]}_{0}) + \delta^2\mathbb{E}[(w_i^\perp)^2|x_i] \\
&\overset{\text{LIE}}{=} \sigma_u^2 + 2\delta\mathbb{E}[w_i\underbrace{\mathbb{E}[u_i|x_i, w_i]}_{0}|x_i] + \delta^2\mathbb{E}[(w_i^\perp)^2|x_i] \\
&= \sigma_u^2 + \delta^2\mathbb{E}[(w_i^\perp)^2|x_i]
\end{aligned}$$

Notice that, by the law of total variance, $\mathbb{E}[(w_i^\perp)^2|x_i] = \text{Var}(w_i^\perp|x_i) + \mathbb{E}[w_i^\perp|x_i]^2$. Now, since $w_i^\perp$ is the component of w_i that is orthogonal to x_i and by the projection interpretation of the conditional variance, $\mathbb{E}[w_i^\perp|x_i] = 0$. And thus, since by assumption $\text{Var}(w_i^\perp|x_i) = \text{Var}(w_i^\perp)$,

$$\begin{aligned}
\mathbb{E}[(w_i^\perp)^2|x_i] &= \text{Var}(w_i^\perp|x_i) \\
&= \text{Var}(w_i^\perp) \\
&= \mathbb{E}[(w_i^\perp)^2] - \mathbb{E}[w_i^\perp]^2 \\
&= \mathbb{E}\left[\left(w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\right)^2\right] - \left(\underbrace{\mathbb{E}[w_i]}_{0} + \frac{\gamma\sigma_w^2}{\sigma_x^2}\underbrace{\mathbb{E}[x_i - \mu_x]}_{0}\right)^2 \\
&= \underbrace{\mathbb{E}[w_i^2]}_{\sigma_w^2} - 2\frac{\gamma\sigma_w^2}{\sigma_x^2}\left(\underbrace{\mathbb{E}[x_i w_i]}_{\gamma\sigma_w^2} - \mu_x \underbrace{\mathbb{E}[w_i]}_{0}\right) + \frac{\gamma^2\sigma_w^4}{\sigma_x^4}\underbrace{\mathbb{E}[(x_i - \mu_x)^2]}_{\sigma_x^2} \\
&= \sigma_w^2 \left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)
\end{aligned}$$

Note that this variance is well defined (positive) only if $\sigma_x^2 \geq \gamma^2\sigma_w^2$. Under this condition,

$$\mathbb{E}[u_{\text{OVB},i}^2|\mathrm{x}_i] = \sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right) \tag{9}$$

Thus, under our set of assumptions, $\mathbb{E}[u_{\text{OVB},i}^2|\mathrm{x}_i]$ does not depend on x_i and $\mathbb{E}[u_{\text{OVB},i}^2|\mathrm{x}_i] = \mathbb{E}[u_{\text{OVB},i}^2]$. We denote this quantity $\sigma_{u_{\text{OVB}}}^2$.

We can now compute the variance of the estimator $\hat{\boldsymbol{\beta}}_{\text{OVB}}$, noting that $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' u_{\text{OVB},i}^2] = \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\mathbb{E}[u_{\text{OVB},i}^2|\mathrm{x}_i]] = \mathbb{E}[\mathrm{x}_i\mathrm{x}_i'\sigma_{u_{\text{OVB}}}^2] = \sigma_{u_{\text{OVB}}}^2\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$. And thus $\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i\mathrm{x}_i' u_{\text{OVB},i}^2]\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']^{-1} = \sigma_{u_{\text{OVB}}}^2\mathbb{E}[\mathrm{x}_i\mathrm{x}_i']$.

Plugin this and equation 8 into equation 7, we get, for $\hat{\beta}_{\text{OVB}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{OVB}}$:

$$\sqrt{n}(\hat{\beta}_{\text{OVB}} - \beta_1) \xrightarrow{d} \mathcal{N}\left(\frac{\delta\gamma\sigma_w^2}{\sigma_x^2}, \ \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{\sigma_x^2}\right)$$

Then, noting that $\rho_{xw} = \text{corr}(x, w) = \frac{\text{cov}(\mu_x + \gamma w + \epsilon, w)}{\sigma_x\sigma_w} = \frac{\gamma\sigma_w}{\sigma_x}$, we have:

$$\sigma_{\text{OVB}}^2 = \text{avar}\left(\hat{\beta}_{\text{OVB}}\right) = \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \rho_{xw}^2\right)}{\sigma_x^2}$$

□

I. 3 Asymptotic distribution of $\hat{\beta}_{\text{CTRL}}$ (Lemma 3)

Proof. The proof is the well know proof of the asymptotic distribution of the OLS. I simply compute $\mathbb{E}[x_{w,i}x'_{w,i}]^{-1}$ to retrieve the variance of the parameter of interest $\boldsymbol{\beta}_{\text{CTRL}}$. We know that we have:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{CTRL}} - \boldsymbol{\beta}_{\text{CTRL}}) \xrightarrow{d} \mathcal{N}\left(0, \mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}]^{-1}\sigma_u^2\right)$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{CTRL}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}]^{-1}$.

$$\mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}] = \mathbb{E}\begin{bmatrix} 1 & x_i & w \\ x_i & x_i^2 & x_i w_i \\ w_i & x_i w_i & w_i^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_x & 0 \\ \mu_x & \sigma_x^2 + \mu_x^2 & \gamma\sigma_w^2 \\ 0 & \gamma\sigma_w^2 & \sigma_w^2 \end{bmatrix}$$

Note that we have $\mathbb{E}[x_i w_i] = \mathbb{E}[x_i]\underbrace{\mathbb{E}[w_i]}_{0} + \mathrm{cov}(x_i, w_i) = \gamma\underbrace{\mathrm{var}(w_i)}_{\sigma_w^2} + \underbrace{\mathrm{cov}(\epsilon_i, w_i)}_{0} = \gamma\sigma_w^2$.

Now, $\mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}]^{-1} = \dfrac{1}{\det(\mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}])}\ {}^t\mathrm{C}$ with C the comatrix of $\mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}]$. We have:

$$\det(\mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}]) = (\sigma_x^2 + \mu_x^2)\sigma_w^2 - \sigma_w^2\mu_x^2 - \gamma^2\sigma_w^4 = \sigma_w^2(\sigma_x^2 - \gamma^2\sigma_w^2)$$

and the "central" component of C, σ_w^2. Thus the central component of interest of $\mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}]^{-1}$ is $\frac{1}{\sigma_x^2 - \gamma^2\sigma_w^2}$. Therefore, for $\hat{\beta}_{\text{CTRL}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{CTRL}}$, we have:

$$\sqrt{n}(\hat{\beta}_{\text{CTRL}} - \beta_1) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\sigma_u^2}{\sigma_x^2 - \gamma^2\sigma_w^2}\right) \tag{10}$$

Then, noting that $\rho_{xw} = \mathrm{corr}(x, w) = \frac{\mathrm{cov}(\mu_x + \gamma w + \epsilon, w)}{\sigma_x\sigma_w} = \frac{\gamma\sigma_w}{\sigma_x}$, we have:

$$\sigma_{\text{CTRL}}^2 = \frac{\sigma_u^2}{\sigma_x^2(1 - \rho_{xw}^2)}$$

□

I. 4 Asymptotic distribution of $\hat{\beta}_{\text{IV}}$ (Lemma 4)

Proof. Since $u_{\text{IV}} = u_{\text{OVB}} = \delta w + u$, we have $\sigma_{u_{\text{IV}}}^2 = \sigma_u^2 + \delta^2\sigma_w^2$. Thus, the usual asymptotic distribution of the IV gives:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{IV}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}\left(0, (\sigma_u^2 + \delta^2\sigma_w^2)\mathbb{E}[\mathrm{z}_i\mathrm{x}'_i]^{-1}\mathbb{E}[\mathrm{z}_i\mathrm{z}'_i]\left(\mathbb{E}[\mathrm{z}_i\mathrm{x}'_i]^{-1}\right)'\right)$$

Note that, when the instruments are weak, the convergence of the sampling distribution to this asymptotic normal distribution does not hold. However, the resulting distribution having fatter tails than the normal and significant estimates from underpowered studies being located in the tails of the distribution, the normal distribution will yield a lower bound for the exaggeration.

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{IV}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{z}_i\mathrm{z}_i]$, $\mathbb{E}[\mathrm{x}_i\mathrm{z}_i']^{-1}$ and its transpose.

$$\mathbb{E}[\mathrm{z}_i\mathrm{z}_i'] = \mathbb{E}\begin{bmatrix}1 & z_i \\ z_i & z_i^2\end{bmatrix} = \begin{bmatrix}1 & \mu_z \\ \mu_z & \sigma_z^2 + \mu_z^2\end{bmatrix}$$

$$\mathbb{E}[\mathrm{z}_i\mathrm{x}_i'] = \begin{bmatrix}1 & \mathbb{E}[x_i] \\ \mathbb{E}[z_i] & \mathbb{E}[z_i x_i]\end{bmatrix} = \begin{bmatrix}1 & \pi_0 + \pi_1\mathbb{E}[z_i] + \gamma\underbrace{\mathbb{E}[w_i]}_{0} + \underbrace{\mathbb{E}[e_i]}_{0} \\ \mu_z & \pi_0\mathbb{E}[z_i] + \pi_1\mathbb{E}[z_i^2] + \gamma\underbrace{\mathbb{E}[z_i w_i]}_{0} + \underbrace{\mathbb{E}[z_i e_i]}_{0}\end{bmatrix} = \begin{bmatrix}1 & \pi_0 + \pi_1\mu_z \\ \mu_z & \pi_0\mu_z + \pi_1(\sigma_z^2 + \mu_z^2)\end{bmatrix}$$

$$\Rightarrow \quad \mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1} = \frac{1}{\pi_1\sigma_z^2}\begin{bmatrix}\pi_0\mu_z + \pi_1(\sigma_z^2 + \mu_z^2) & -\pi_0 - \pi_1\mu_z \\ -\mu_z & 1\end{bmatrix}$$

Thus,

$$\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']\left(\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\right)' = \frac{1}{\pi_1\sigma_z^2}\begin{bmatrix}2\pi_0\mu_z + \pi_1(\sigma_z^2 + \mu_z^2) + \frac{\pi_0^2}{\pi_1} & -\mu_z - \frac{\pi_0}{\pi_1} \\ -\mu_z - \frac{\pi_0}{\pi_1} & \frac{1}{\pi_1}\end{bmatrix}$$

And so, for $\hat{\beta}_{\text{IV}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{IV}}$, we have:

$$\sqrt{n}\left(\hat{\beta}_{\text{IV}} - \beta_1\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\sigma_u^2 + \delta^2\sigma_w^2}{\pi_1^2\sigma_z^2}\right) \tag{11}$$

Now, since $\rho_{xz} = \text{corr}(x_i, z_i) = \frac{\text{cov}(\pi_0 + \pi_1 z_i + \gamma w_i + e_i, z_i)}{\sigma_x\sigma_z} = \pi_1\frac{\sigma_z}{\sigma_x}$,

$$\sqrt{n}\left(\hat{\beta}_{\text{IV}} - \beta_1\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\sigma_u^2 + \delta^2\sigma_w^2}{\sigma_x^2\rho_{xz}^2}\right)$$

□

I. 5 Asymptotic distribution of $\hat{\beta}_{\text{RED}}$ (Lemma 5)

Proof. The proof is straightforward: this is the usual bivariate, unbiased case, with and error term equal to $(\delta + \beta_1\gamma)w_i + u_i + \beta_1 e_i$. Since w, u and ϵ_{RED} uncorrelated, its variance is $(\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_e^2$. □